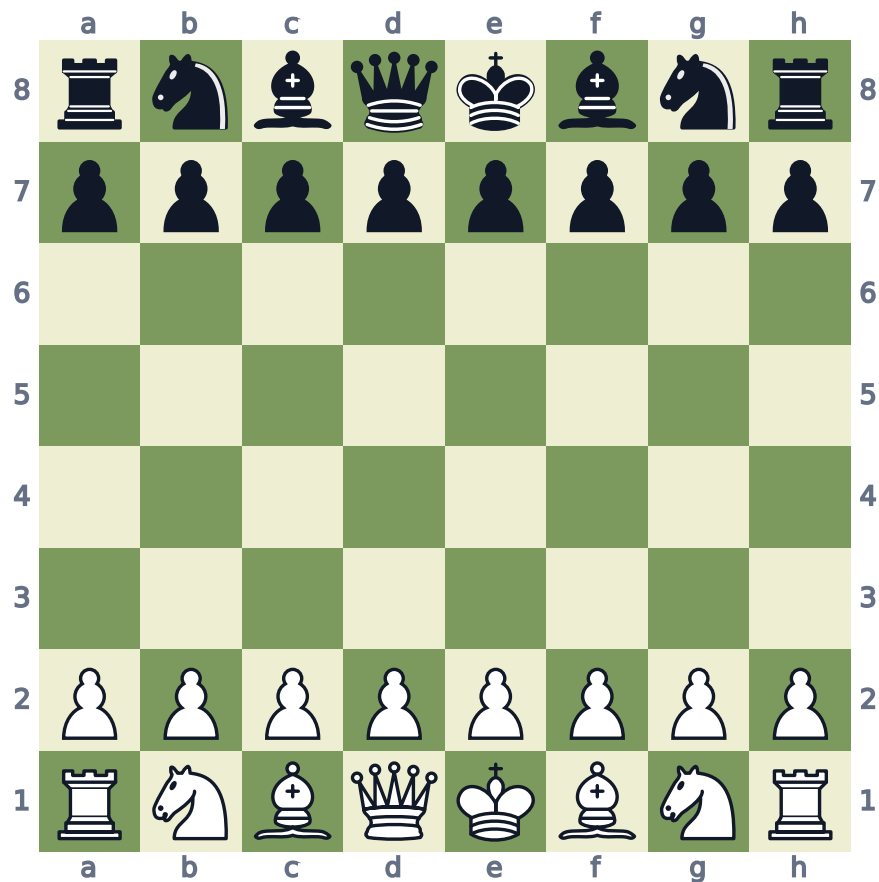


Essential Endgames

Exact foundations and practical technical play



Master the positions that must be known, while learning how to calculate unfamiliar endings from king activity, pawn races, checking geometry, and counterplay.

55 learning units | explanations | board laboratories | review system

ORIENTATION

What this course will change

COURSE PROMISE

Master the positions that must be known, while learning how to calculate unfamiliar endings from king activity, pawn races, checking geometry, and counterplay.

YOU WILL LEARN TO

- Use opposition and key squares
- Calculate pawn races exactly
- Handle minor-piece endings
- Apply rook and queen endgame techniques
- Recognize drawing resources and fortresses

HOW TO USE THE COURSE

- Study one learning unit at a time. Say the rule in your own words before continuing.
- At each diagram, hide the model answer. Write candidates and the opponent's best reply.
- Use the rule during a slow game, then record whether it actually changed a decision.
- Review after 1, 3, 7, 14, and 30 days. Retrieval matters more than rereading.

SCOPE

This course contains 55 integrated learning units. Principles guide attention, but tactics and exact calculation always decide the move.

CORE METHOD

The decision protocol

USE THIS EVERY TIME

A reliable process is more valuable than a brilliant idea used occasionally. Practice the sequence until it becomes automatic under clock pressure.

1**Count material and passed pawns.****2****Find the most active king and piece.****3****Identify the exact target position.****4****Calculate exchanges and pawn races completely.****5****Remove counterplay before making routine progress.****STOPPING RULE**

Choose when one candidate is sufficiently verified for the clock situation. Before moving, inspect the position once from the opponent's perspective.

COURSE MAP

Learning units

- 01 Activate the King
- 02 Put the Rook Behind a Passed Pawn
- 03 Use the Opposition
- 04 Create an Outside Passed Pawn
- 05 Create a Distant Passed Pawn Before Cashing It In
- 06 Centralize Before Starting a Pawn Race
- 07 Do Not Rush Pawn Moves
- 08 Trade Pieces, Not Pawns, When Ahead
- 09 Choose Rook Activity Over Pawn Grabbing
- 10 Calculate Pawn Races Exactly
- 11 Knights Struggle Against Distant Passed Pawns
- 12 Bishops Gain Value on Both Wings
- 13 Opposite-Colored Bishops Favor the Defender
- 14 Opposite-Colored Bishops Strengthen Middlegame Attacks
- 15 Fix Pawns on the Color Your Bishop Can Attack
- 16 Put Your Pawns Opposite Your Own Bishop's Color
- 17 A Knight Needs an Active King
- 18 Use Triangulation to Lose a Tempo
- 19 Create Zugzwang by Removing Useful Moves
- 20 Sacrifice a Minor Piece for the Last Pawns When It Draws

COURSE MAP

Learning units

- 21 Cut Off the Enemy King With the Rook
- 22 Build the Lucena Bridge
- 23 Use the Philidor Third-Rank Defense
- 24 Check From the Side With Adequate Distance
- 25 Know the Wrong-Bishop Rook-Pawn Draw
- 26 Centralize the Queen With Tempo
- 27 King Safety Dominates Queen Endgames
- 28 Build a Perpetual-Check Map
- 29 Calculate Queen Trades to the Final Pawn Ending
- 30 Connected Passed Pawns Can Outweigh a Rook
- 31 Win the Key Squares
- 32 Use the Rule of the Square Precisely
- 33 Map Corresponding Squares
- 34 Preserve a Reserve Tempo
- 35 Calculate the Pawn Breakthrough
- 36 Shoulder the Enemy King Away
- 37 Use Distant Opposition
- 38 Know the Knight's Rook-Pawn Problem
- 39 Check the Bishop's Promotion Color
- 40 Use the Outside Passer as a Decoy

COURSE MAP

Learning units

- 41 Use a Protected Passer as a Barrier
- 42 Respect Connected Passers on the Sixth
- 43 Give Rook Checks from Maximum Distance
- 44 Use the Seventh Rank as an Endgame Base
- 45 Learn the Vancura Defense
- 46 Defend from the Short Side
- 47 Cut the King Off Before Pushing
- 48 Verify the Pawn Ending Before Trading Rooks
- 49 Centralize the Queen with Check
- 50 Control the Promotion Square in Queen versus Pawn
- 51 Identify Perpetual-Check Zones
- 52 Trade Queens with a Cross-Check
- 53 Build the Fortress Around One Entry Point
- 54 Coordinate Bishop and Knight for Mate
- 55 Extinguish Counterplay Before Converting

LEARNING UNIT 01

Activate the King

THE RULE

Use the king as a fighting piece once mating danger and queens are reduced.

WHY IT WORKS

The king attacks pawns, supports passers, controls entry squares, and often outweighs a distant minor piece in local battles.

RECOGNITION CUES

Queens traded; few pieces remain; king can approach the center safely.

APPLY IT OVER THE BOARD

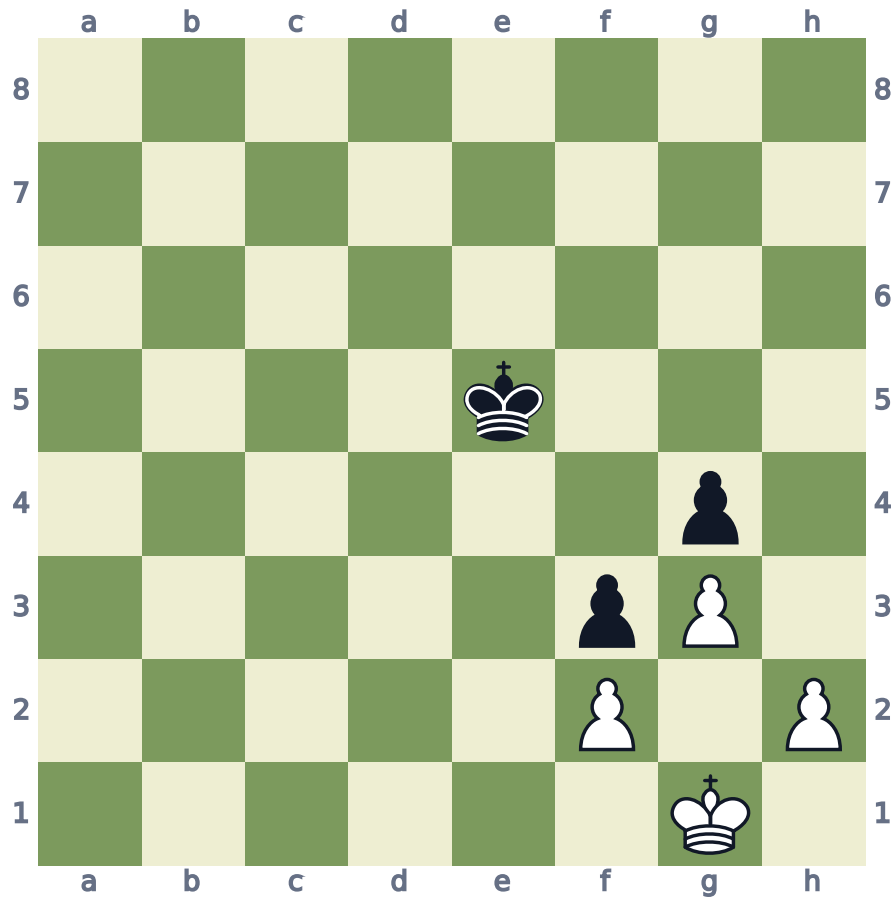
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not centralize into tactical checks, mating nets, or a lost pawn race.

BOARD LABORATORY 01

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Activate the King'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The king attacks pawns, supports passers, controls entry squares, and often outweighs a distant minor piece in local battles. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

In ten endings, find the shortest legal king route to the most important central square.

LEARNING UNIT 02

Put the Rook Behind a Passed Pawn

THE RULE

Place a rook behind a passed pawn, whether it is yours or the opponent's, when the file is accessible.

WHY IT WORKS

From behind, the rook's checking and pushing distance grows as the pawn advances. In front, the rook becomes increasingly restricted.

RECOGNITION CUES

Open file behind a passer; rook ending; pawn can advance with support.

APPLY IT OVER THE BOARD

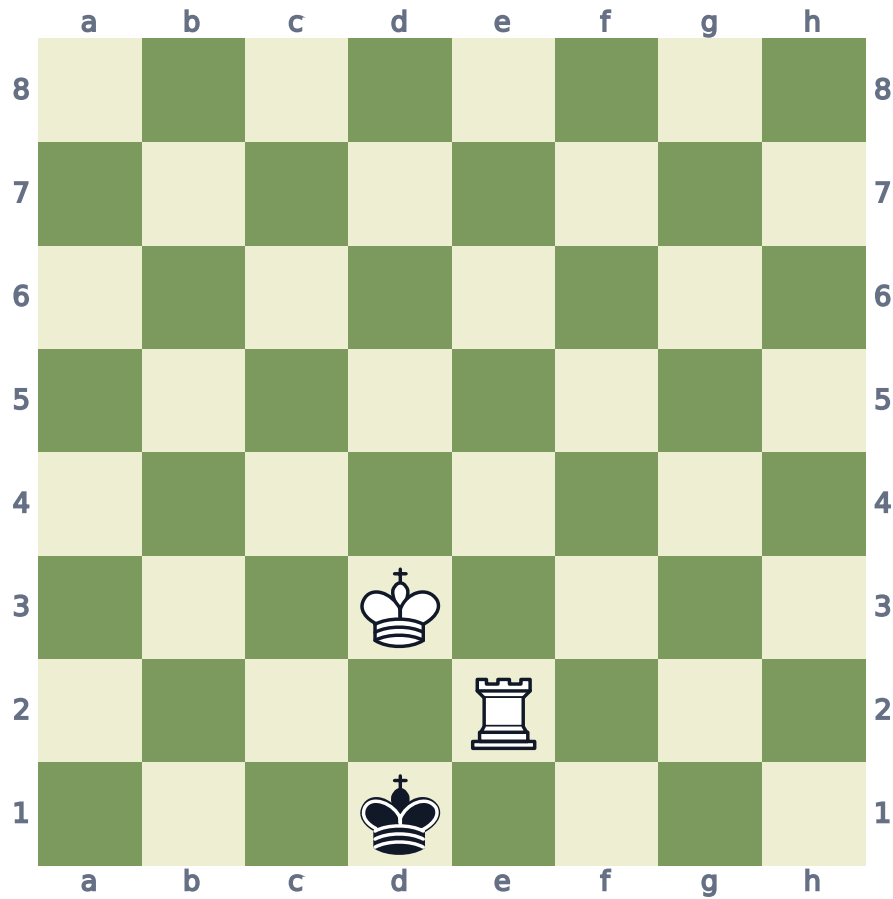
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Side checks, cutting off the king, or immediate tactical activity can be more important than ideal placement.

BOARD LABORATORY 02

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Put the Rook Behind a Passed Pawn'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: From behind, the rook's checking and pushing distance grows as the pawn advances. In front, the rook becomes increasingly restricted. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up five rook-and-passer positions and compare rook behind, beside, and in front.

LEARNING UNIT 03

Use the Opposition

THE RULE

In king-and-pawn endings, arrange kings so the opponent must yield the critical entry square.

WHY IT WORKS

Kings cannot stand adjacent. Move parity and opposition convert this rule into controlled access or exclusion.

RECOGNITION CUES

Bare kings and pawns; direct, distant, or diagonal opposition; a key square decides promotion.

APPLY IT OVER THE BOARD

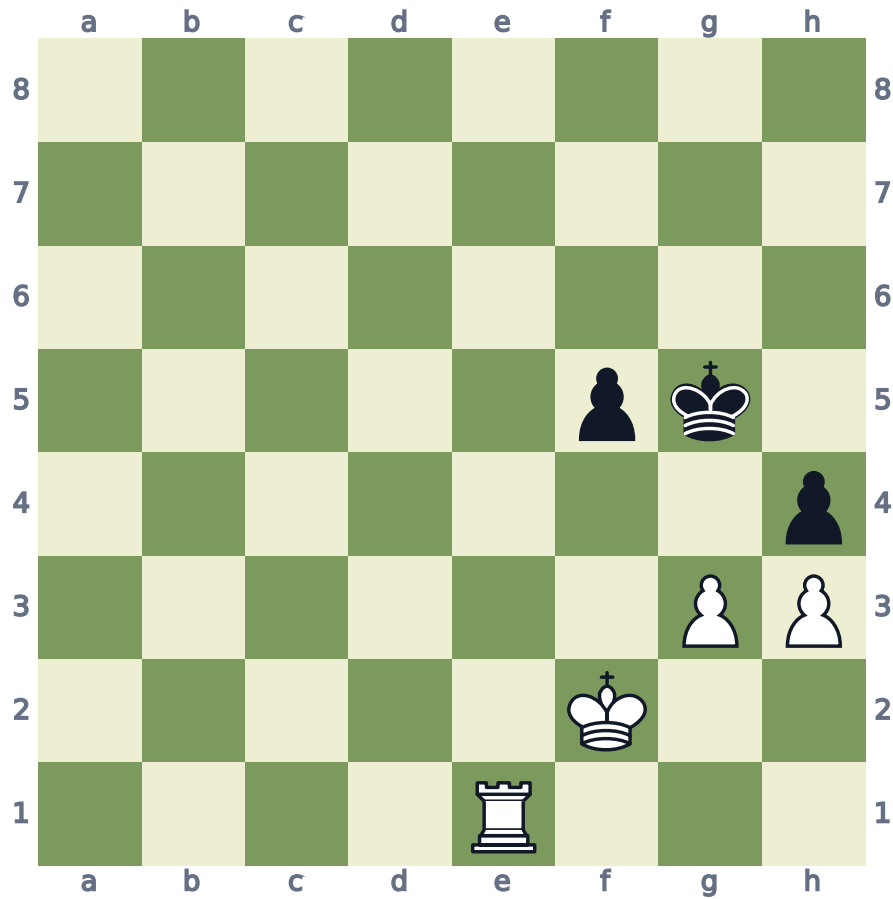
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Opposition is a tool, not the objective. Sometimes yielding it gains the key square by triangulation.

BOARD LABORATORY 03

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Use the Opposition'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Kings cannot stand adjacent. Move parity and opposition convert this rule into controlled access or exclusion. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten king-and-pawn positions from both sides until you can name the key square before moving.

LEARNING UNIT 04

Create an Outside Passed Pawn

THE RULE

Create a passer far from the main pawn mass to divert the enemy king.

WHY IT WORKS

The distant threat stretches the defender. While the king stops it, your king can invade and collect pawns elsewhere.

RECOGNITION CUES

Pawn majorities on opposite wings; kings can reach only one side quickly; exchanges can create a remote passer.

APPLY IT OVER THE BOARD

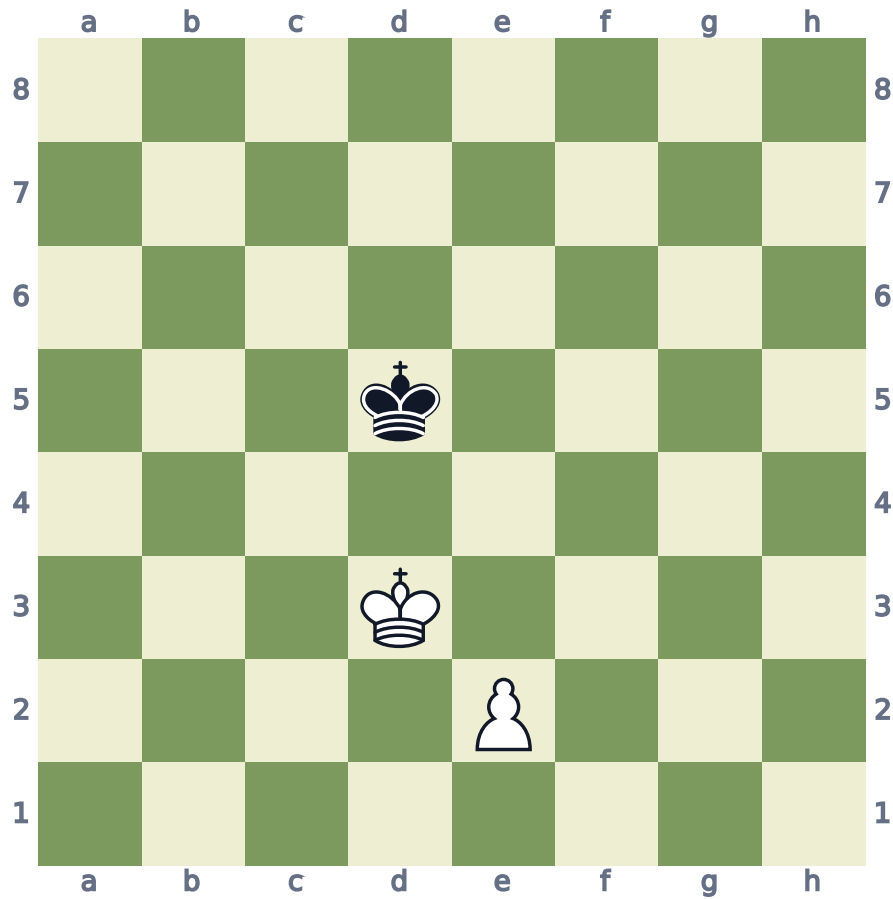
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

An outside passer may be too weak if enemy pieces stop it without abandoning the other wing.

BOARD LABORATORY 04

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Create an Outside Passed Pawn'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The distant threat stretches the defender. While the king stops it, your king can invade and collect pawns elsewhere. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

For five pawn endings, calculate whether creating the outside passer wins after the forced king route.

LEARNING UNIT 05

Create a Distant Passed Pawn Before Cashing It In

THE RULE

Use the threat of promotion to gain king activity or win pawns rather than pushing automatically.

WHY IT WORKS

A passed pawn's greatest value may be distraction. Keeping it alive can tie the enemy king or rook more effectively than sacrificing it immediately.

RECOGNITION CUES

Defender is tied to the passer; your king can enter elsewhere; the pawn need not advance yet.

APPLY IT OVER THE BOARD

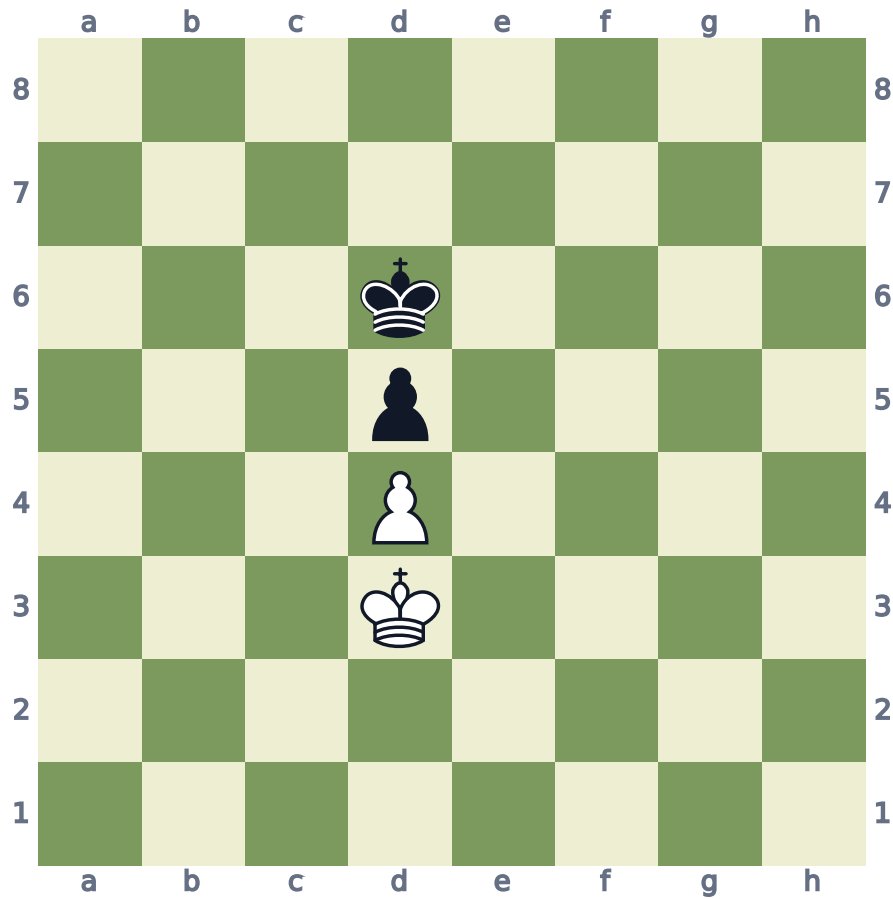
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Push when the advance gains tempo, reaches safety, or forces a decisive concession.

BOARD LABORATORY 05

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Create a Distant Passed Pawn Before Cashing It In'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A passed pawn's greatest value may be distraction. Keeping it alive can tie the enemy king or rook more effectively than sacrificing it immediately. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Study ten positions and compare the passer's value as a moving threat versus a fixed restraint.

LEARNING UNIT 06

Centralize Before Starting a Pawn Race

THE RULE

Improve the king and pieces before committing pawns to a race when no immediate breakthrough is required.

WHY IT WORKS

Advanced pawns become targets and lose flexibility. An active king can support them while attacking the opponent's base.

RECOGNITION CUES

Both wings have pawns; no forced promotion; kings are distant from the center.

APPLY IT OVER THE BOARD

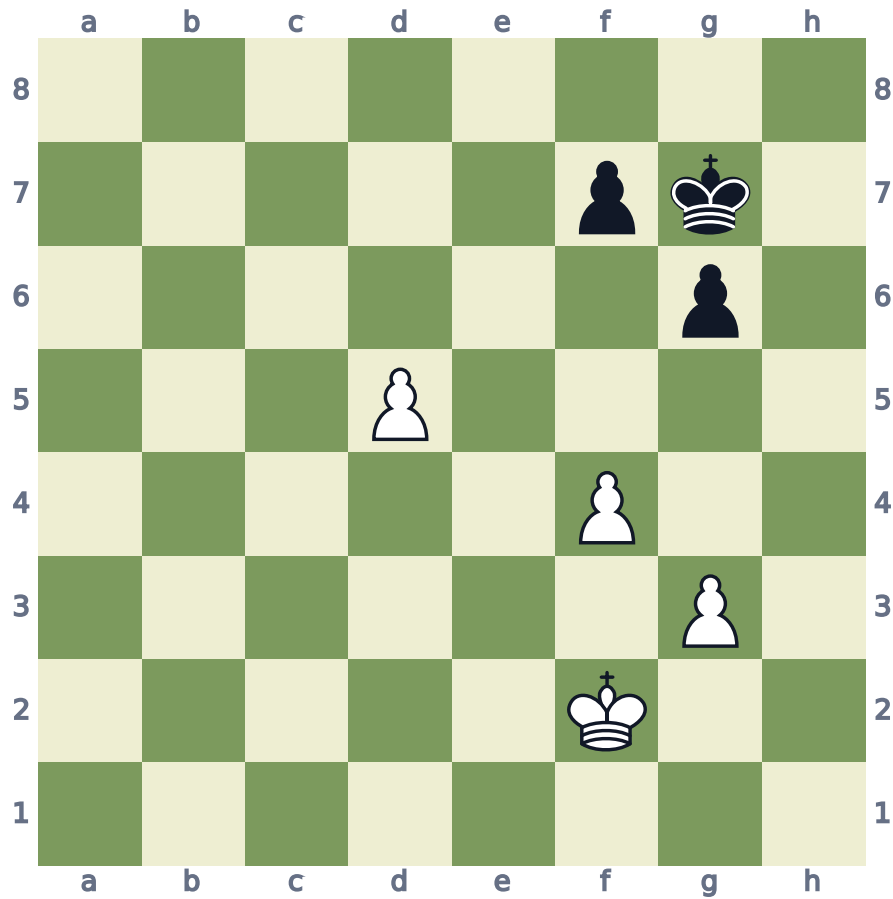
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

If one side has a calculable winning race, delaying may throw away the tempo that makes it work.

BOARD LABORATORY 06

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Centralize Before Starting a Pawn Race'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Advanced pawns become targets and lose flexibility. An active king can support them while attacking the opponent's base. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

For ten pawn races, calculate both immediate advance and one king-improving move first.

LEARNING UNIT 07

Do Not Rush Pawn Moves

THE RULE

Preserve pawn tempi and structure until the move creates a concrete gain.

WHY IT WORKS

Pawn moves can lose opposition, create targets, and exhaust reserve tempi. In endings, one tempo often determines promotion.

RECOGNITION CUES

Few pieces; symmetrical pawns; zugzwang or opposition may arise.

APPLY IT OVER THE BOARD

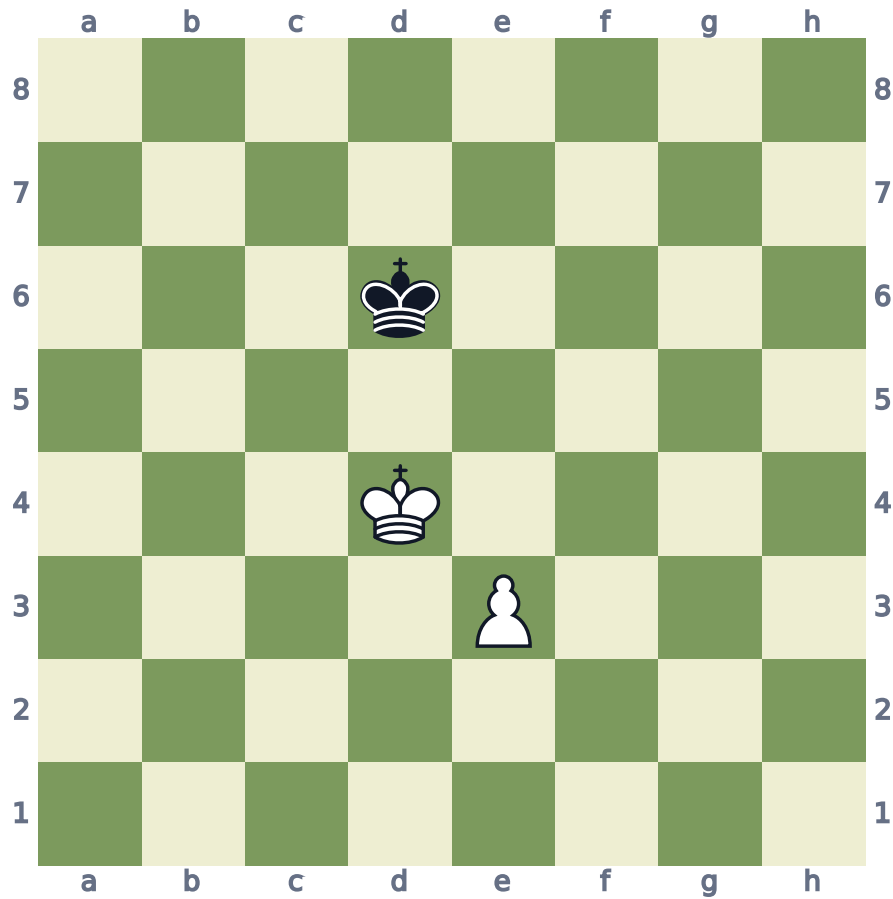
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Push when it fixes a pawn, creates a passer, gains space with safety, or wins the race.

BOARD LABORATORY 07

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Do Not Rush Pawn Moves'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Pawn moves can lose opposition, create targets, and exhaust reserve tempi. In endings, one tempo often determines promotion. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Play ten king-and-pawn endings while recording which pawn moves are reserve tempi.

LEARNING UNIT 08

Trade Pieces, Not Pawns, When Ahead

THE RULE

When materially ahead, exchange active pieces while preserving enough pawns to win.

WHY IT WORKS

Fewer enemy pieces reduce tactical and perpetual-check chances, while remaining pawns provide promotion potential.

RECOGNITION CUES

Extra piece or exchange; safe king; favorable simplification is available.

APPLY IT OVER THE BOARD

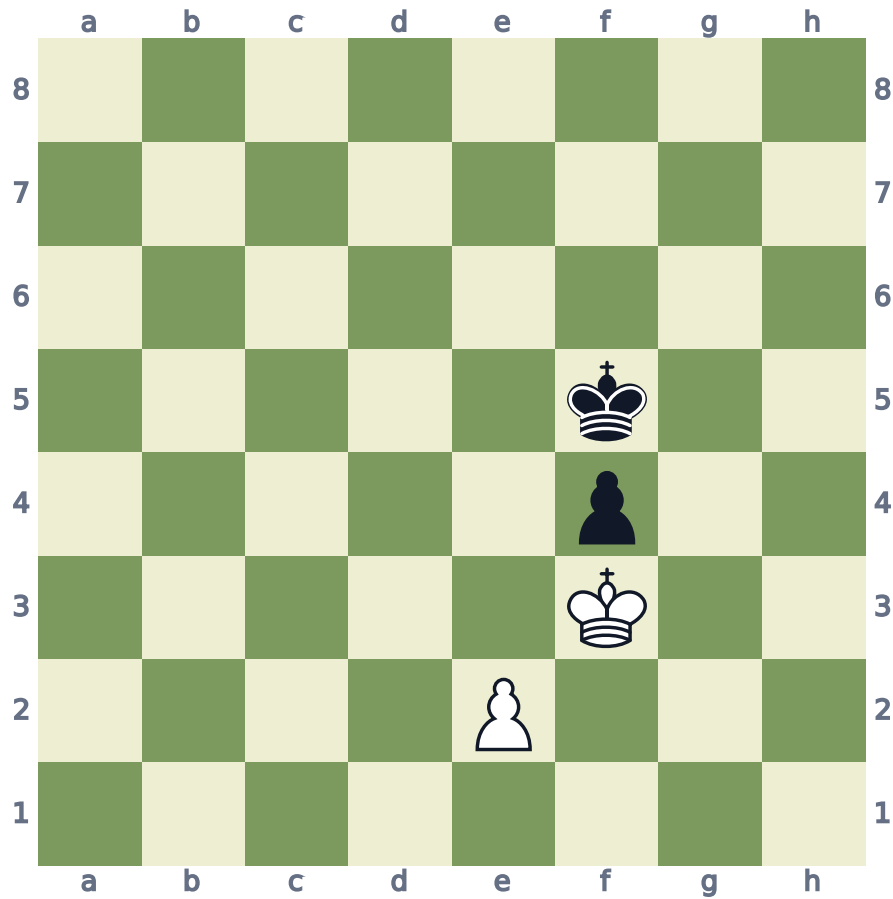
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not trade into a fortress or remove all winning pawns. Sometimes pawn trades eliminate counterplay more effectively.

BOARD LABORATORY 08

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Trade Pieces, Not Pawns, When Ahead'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Fewer enemy pieces reduce tactical and perpetual-check chances, while remaining pawns provide promotion potential. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

For ten winning positions, compare a piece trade and a pawn trade by evaluating the final conversion method.

LEARNING UNIT 09

Choose Rook Activity Over Pawn Grabbing

THE RULE

Prefer checks, king restriction, and penetration when taking a pawn would make the rook passive.

WHY IT WORKS

Active rooks can create perpetual checks, attack several pawns, and support passers. A passive rook may be worth less than the pawn it collected.

RECOGNITION CUES

Enemy pawn is available; rook can invade the seventh rank or cut off the king; king safety matters.

APPLY IT OVER THE BOARD

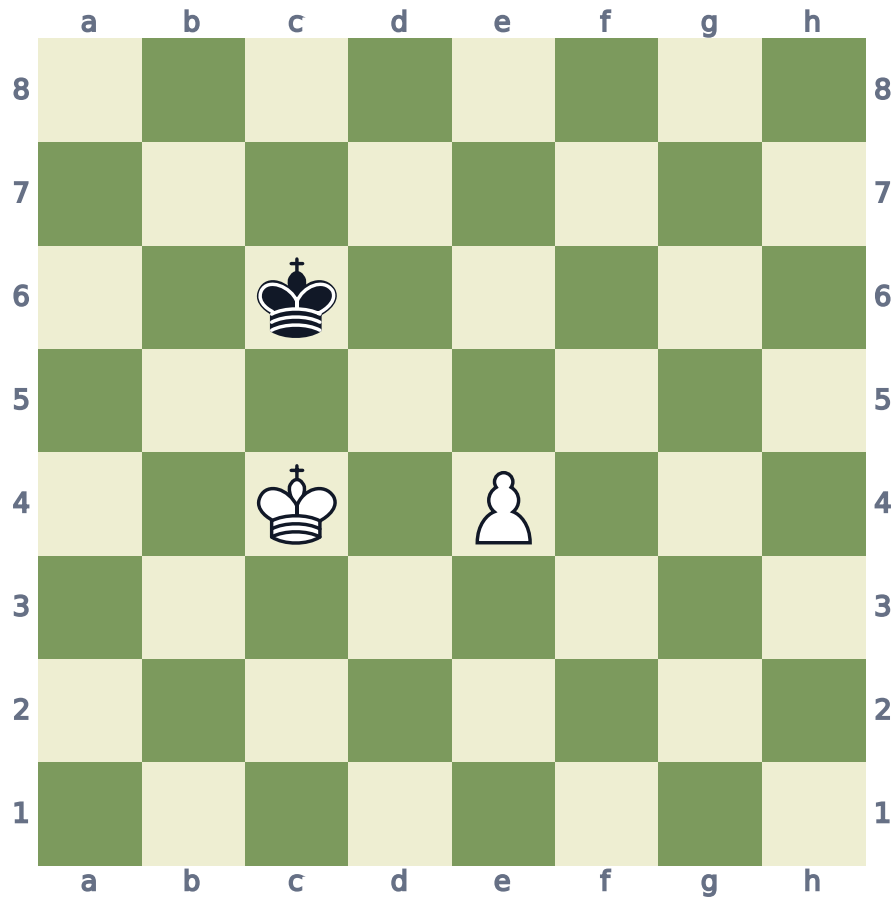
- Count material and passed pawns.
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- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Take the pawn when it is safe, important, and does not surrender the active position.

BOARD LABORATORY 09

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Choose Rook Activity Over Pawn Grabbing'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Active rooks can create perpetual checks, attack several pawns, and support passers. A passive rook may be worth less than the pawn it collected. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

In ten rook endings, calculate one pawn grab and one activity move, then compare checking distance and king restriction.

LEARNING UNIT 10

Calculate Pawn Races Exactly

THE RULE

Count moves to promotion, checks, captures, and king entry rather than relying on visual intuition.

WHY IT WORKS

Pawn races are concrete. Promotion with check, a diagonal king route, or one reserve tempo can reverse the result.

RECOGNITION CUES

Passed pawns on both wings; kings cannot stop everything; exchanges lead to a race.

APPLY IT OVER THE BOARD

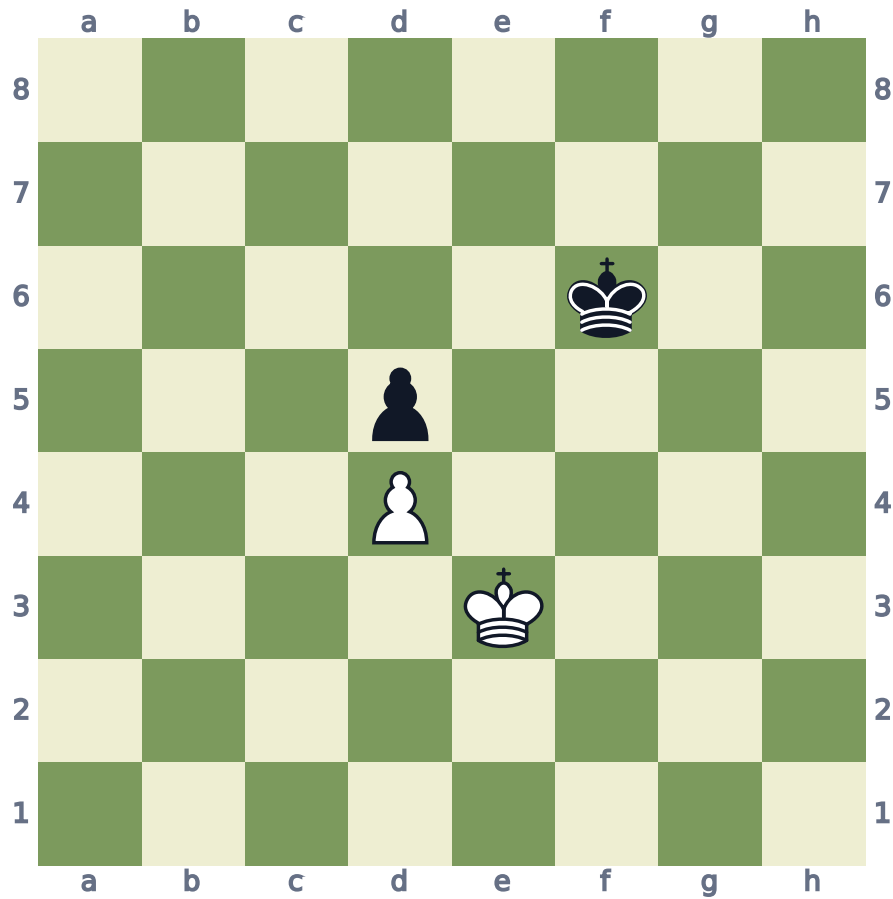
- Count material and passed pawns.
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- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Counting must include tactical details such as promotion square control and whether the new queen gives check.

BOARD LABORATORY 10

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Calculate Pawn Races Exactly'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Pawn races are concrete. Promotion with check, a diagonal king route, or one reserve tempo can reverse the result. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Solve twenty pawn races without moving pieces, then verify on a board and record every counting error.

LEARNING UNIT 11

Knights Struggle Against Distant Passed Pawns

THE RULE

Create or support passers far from an enemy knight's current zone.

WHY IT WORKS

Knights change wings slowly and cannot lose a tempo easily. A distant passer can overload their short-range movement.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

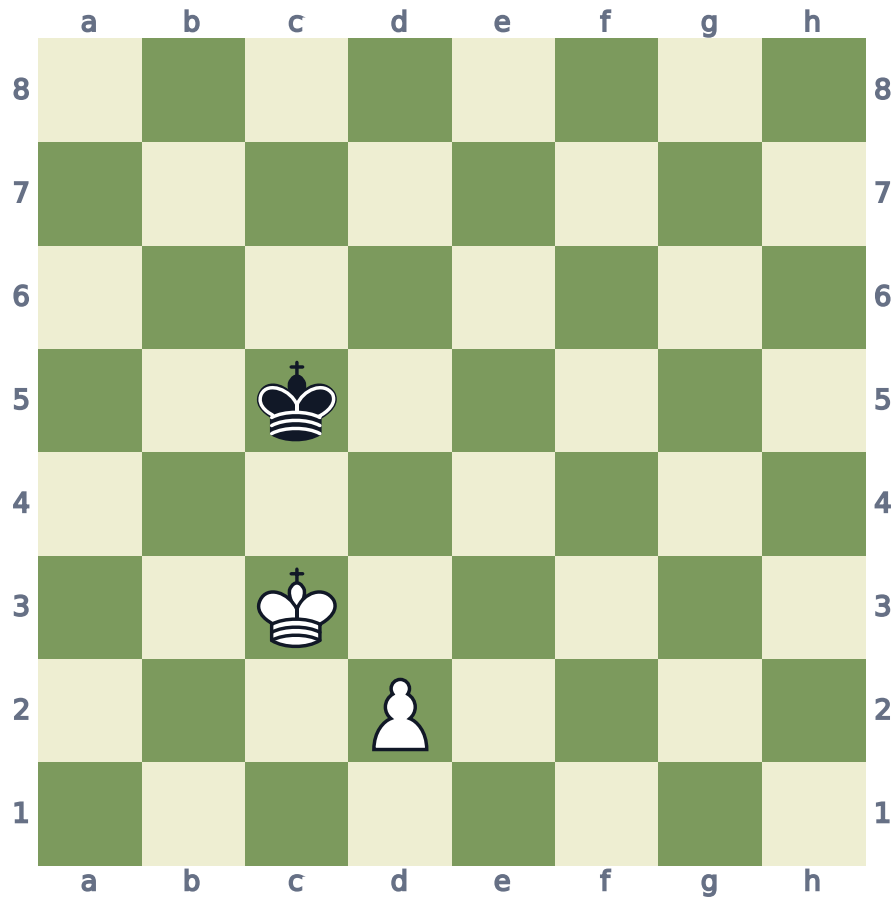
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- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 11

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Knights Struggle Against Distant Passed Pawns'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Knights change wings slowly and cannot lose a tempo easily. A distant passer can overload their short-range movement. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about knights struggle against distant passed pawns. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 12

Bishops Gain Value on Both Wings

THE RULE

Keep play on both flanks when your bishop can switch faster than the opponent's knight or king.

WHY IT WORKS

Long-range movement lets a bishop influence separated targets in one move, increasing the cost of defending two wings.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

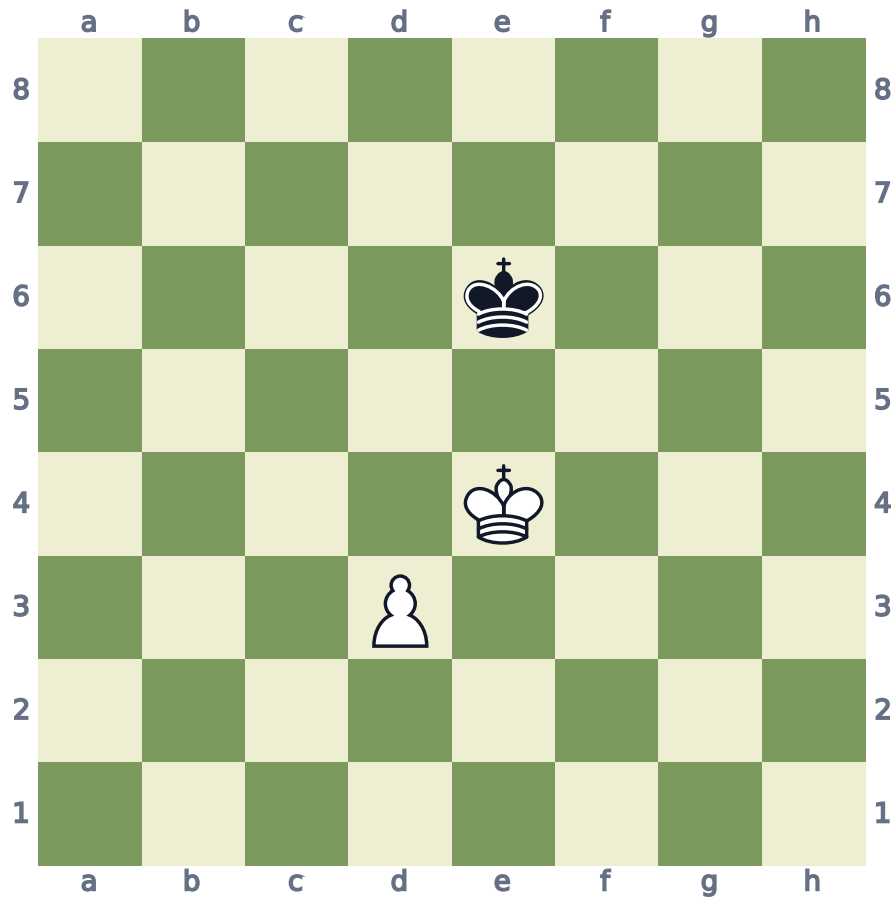
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- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 12

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Bishops Gain Value on Both Wings'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Long-range movement lets a bishop influence separated targets in one move, increasing the cost of defending two wings. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about bishops gain value on both wings. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 13

Opposite-Colored Bishops Favor the Defender

THE RULE

Expect drawing chances when each bishop controls squares the other cannot challenge, especially with pawns on one wing.

WHY IT WORKS

The attacker cannot exchange the defender's bishop or contest its blockade squares. Extra pawns may be insufficient without king entry or connected passers.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

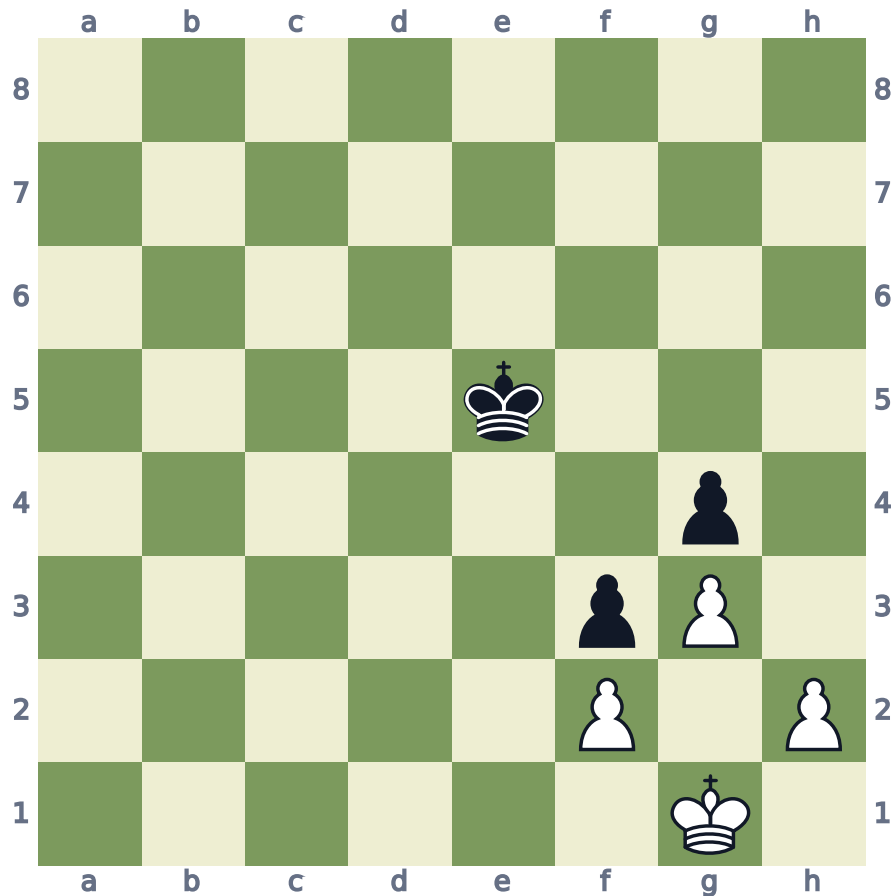
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- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 13

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Opposite-Colored Bishops Favor the Defender'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The attacker cannot exchange the defender's bishop or contest its blockade squares. Extra pawns may be insufficient without king entry or connected passers. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about opposite-colored bishops favor the defender. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 14

Opposite-Colored Bishops Strengthen Middlegame Attacks

THE RULE

With queens or rooks present, attack on the color controlled by your bishop and inaccessible to the opponent's bishop.

WHY IT WORKS

The same color separation that aids defense in pure endings can create an uncontested attacking color complex in the middlegame.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

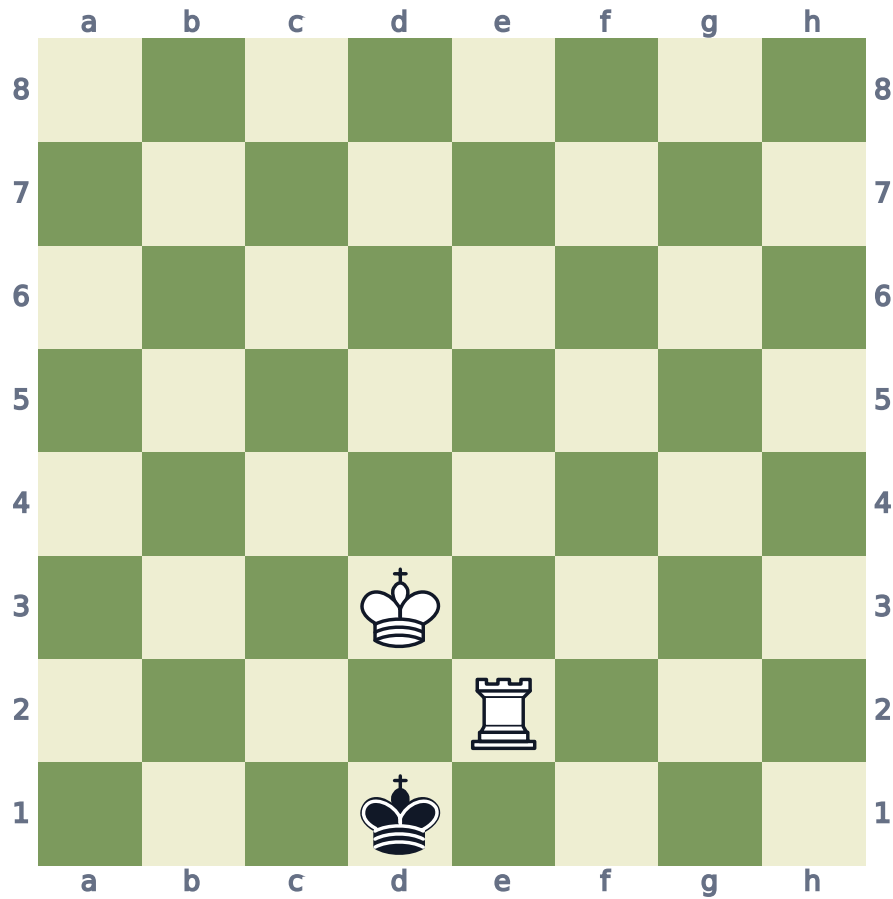
- Count material and passed pawns.
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- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 14

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Opposite-Colored Bishops Strengthen Middlegame Attacks'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The same color separation that aids defense in pure endings can create an uncontested attacking color complex in the middlegame. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about opposite-colored bishops strengthen middlegame attacks. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 15

Fix Pawns on the Color Your Bishop Can Attack

THE RULE

Place or induce enemy pawns onto squares accessible to your bishop before entering the ending.

WHY IT WORKS

Fixed pawns become long-range targets. Your king and bishop can combine against them while the opposing minor piece may lack equivalent access.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

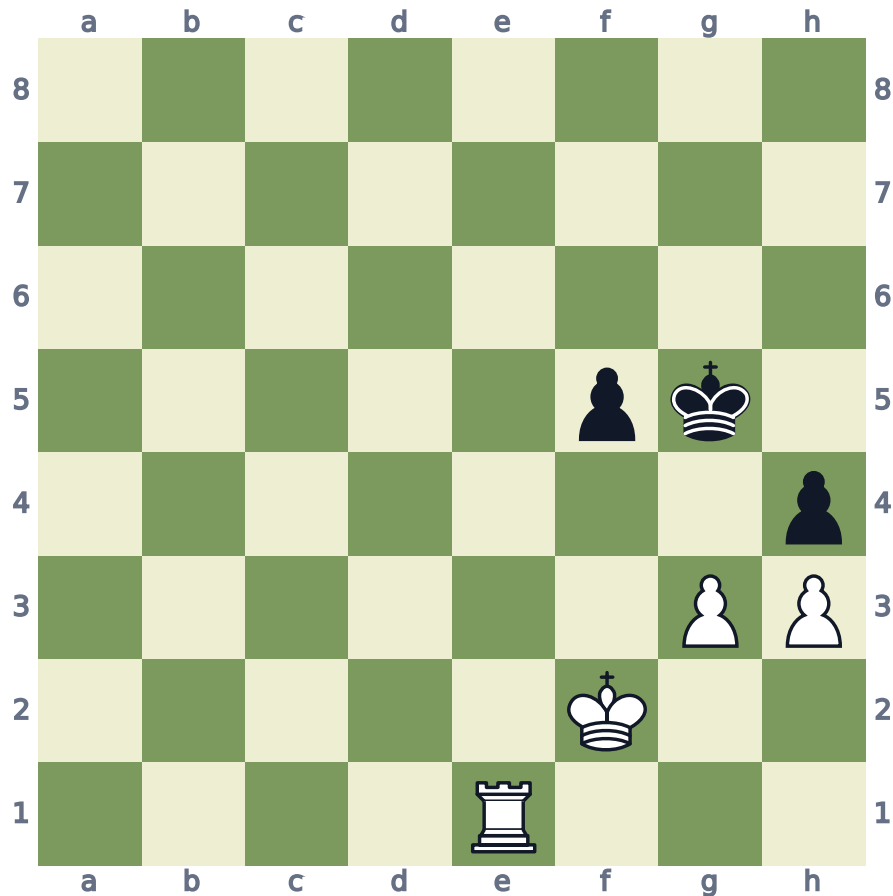
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 15

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Fix Pawns on the Color Your Bishop Can Attack'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Fixed pawns become long-range targets. Your king and bishop can combine against them while the opposing minor piece may lack equivalent access. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about fix pawns on the color your bishop can attack. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 16

Put Your Pawns Opposite Your Own Bishop's Color

THE RULE

When practical, place pawns on squares your bishop does not occupy so they do not block it.

WHY IT WORKS

This division of labor lets pawns control one color and the bishop control the other, maximizing mobility and reducing self-obstruction.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

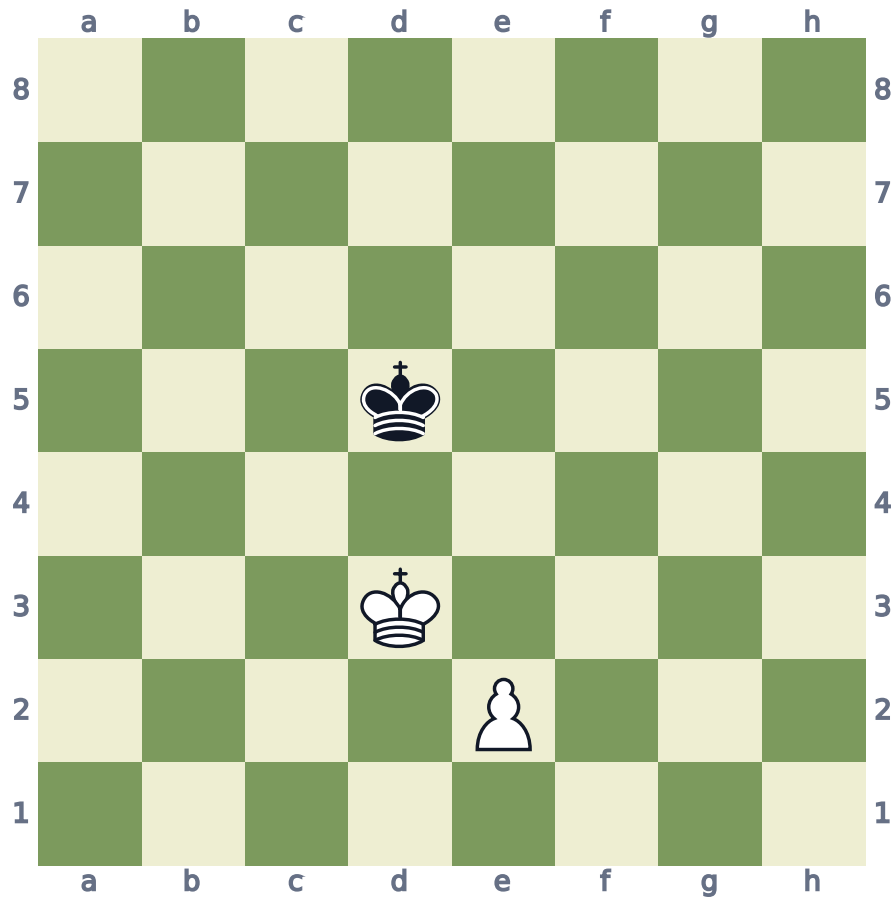
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 16

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Put Your Pawns Opposite Your Own Bishop's Color'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: This division of labor lets pawns control one color and the bishop control the other, maximizing mobility and reducing self-obstruction. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about put your pawns opposite your own bishop's color. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 17

A Knight Needs an Active King

THE RULE

Use the king to protect the knight's outpost and address distant threats.

WHY IT WORKS

The knight's limited range makes king cooperation essential. Together they can create barriers, forks, and zugzwang nets.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

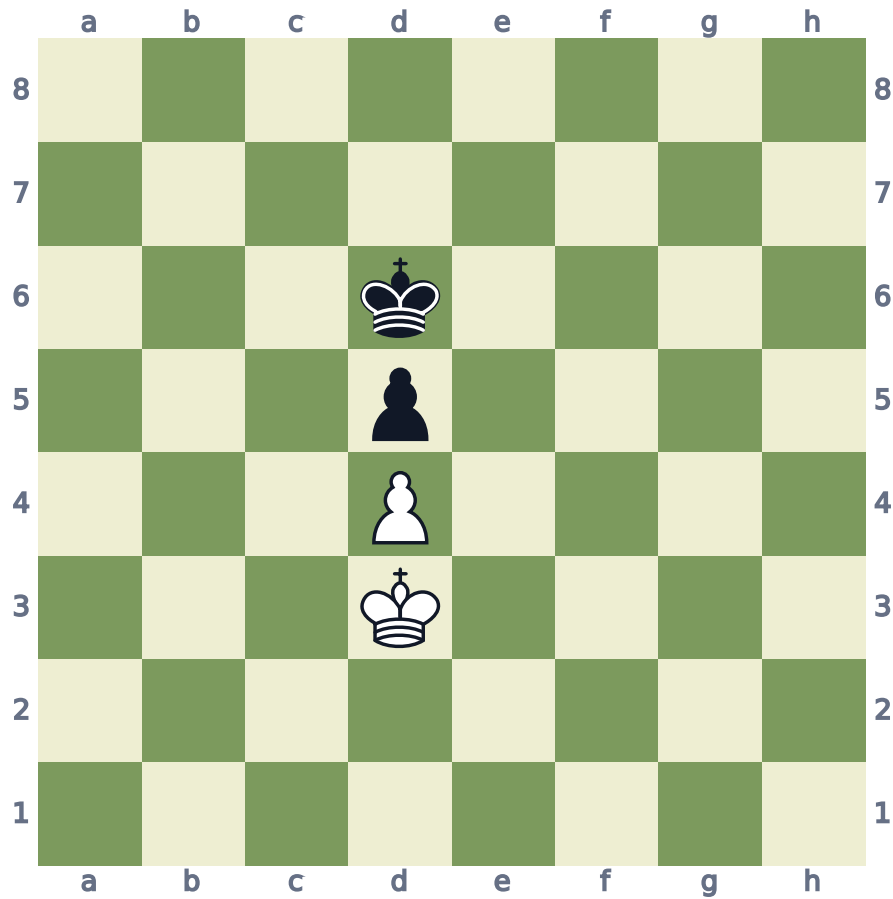
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 17

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'A Knight Needs an Active King'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The knight's limited range makes king cooperation essential. Together they can create barriers, forks, and zugzwang nets. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about a knight needs an active king. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 18

Use Triangulation to Lose a Tempo

THE RULE

Maneuver the king or bishop in a loop to return to the same position with the opponent to move.

WHY IT WORKS

Endgame move parity can force a defender away from a critical square. Triangulation converts spare mobility into zugzwang.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

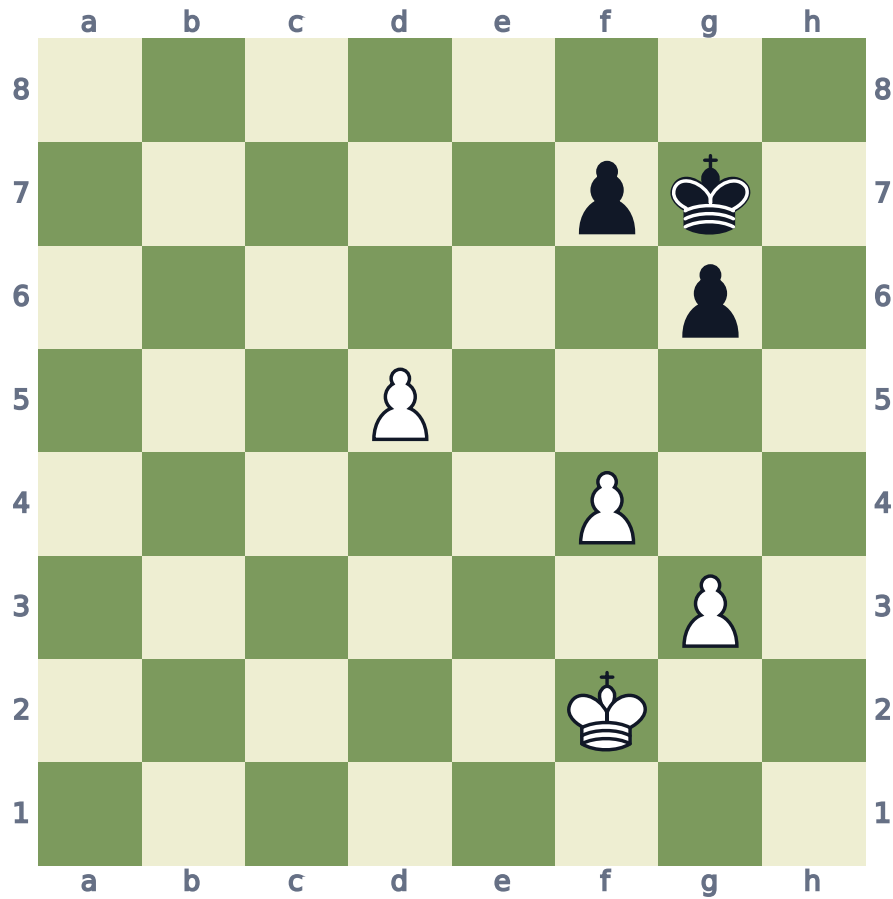
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 18

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Use Triangulation to Lose a Tempo'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Endgame move parity can force a defender away from a critical square. Triangulation converts spare mobility into zugzwang. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about use triangulation to lose a tempo. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 19

Create Zugzwang by Removing Useful Moves

THE RULE

Restrict pawn and piece moves until every legal reply concedes a square or pawn.

WHY IT WORKS

Zugzwang works because passing is illegal. Restraint turns the opponent's move obligation into a weakness.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

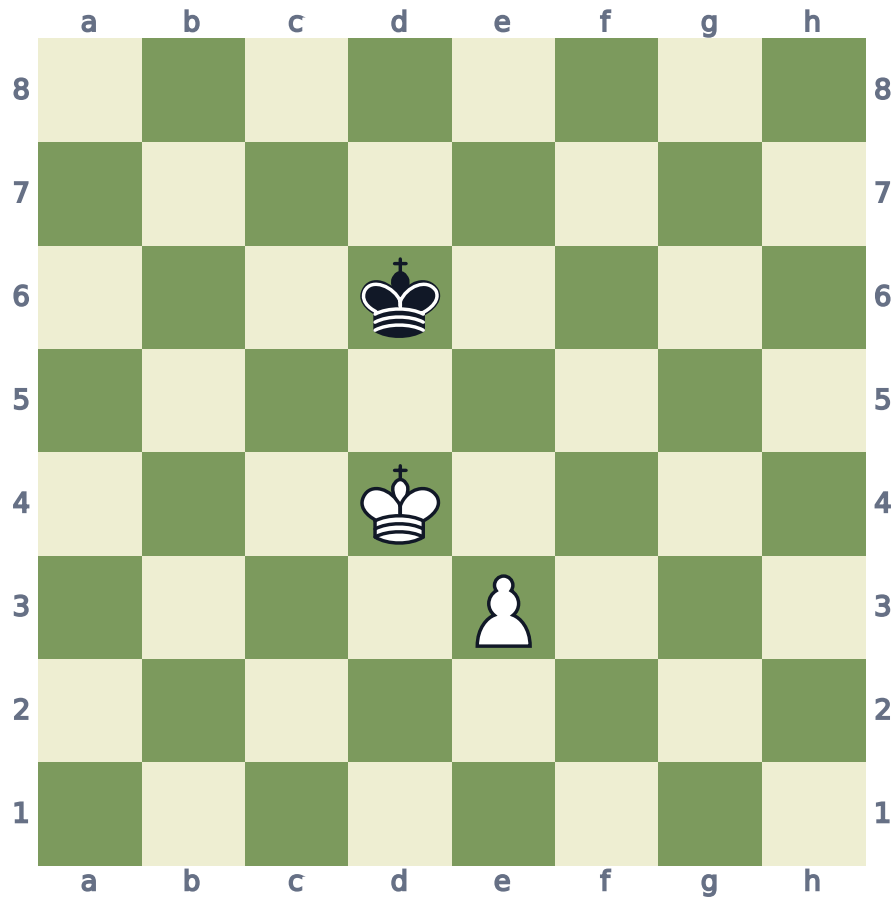
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 19

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Create Zugzwang by Removing Useful Moves'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Zugzwang works because passing is illegal. Restraint turns the opponent's move obligation into a weakness. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about create zugzwang by removing useful moves. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 20

Sacrifice a Minor Piece for the Last Pawns When It Draws

THE RULE

Give the piece for all remaining enemy pawns if the opponent cannot mate with the material left.

WHY IT WORKS

Material value depends on winning capacity. Eliminating every pawn may create an immediate theoretical draw despite losing the piece.

RECOGNITION CUES

Queens and rooks are gone or can be exchanged; pawn color and wing distribution matter; king routes and piece limitations dominate

APPLY IT OVER THE BOARD

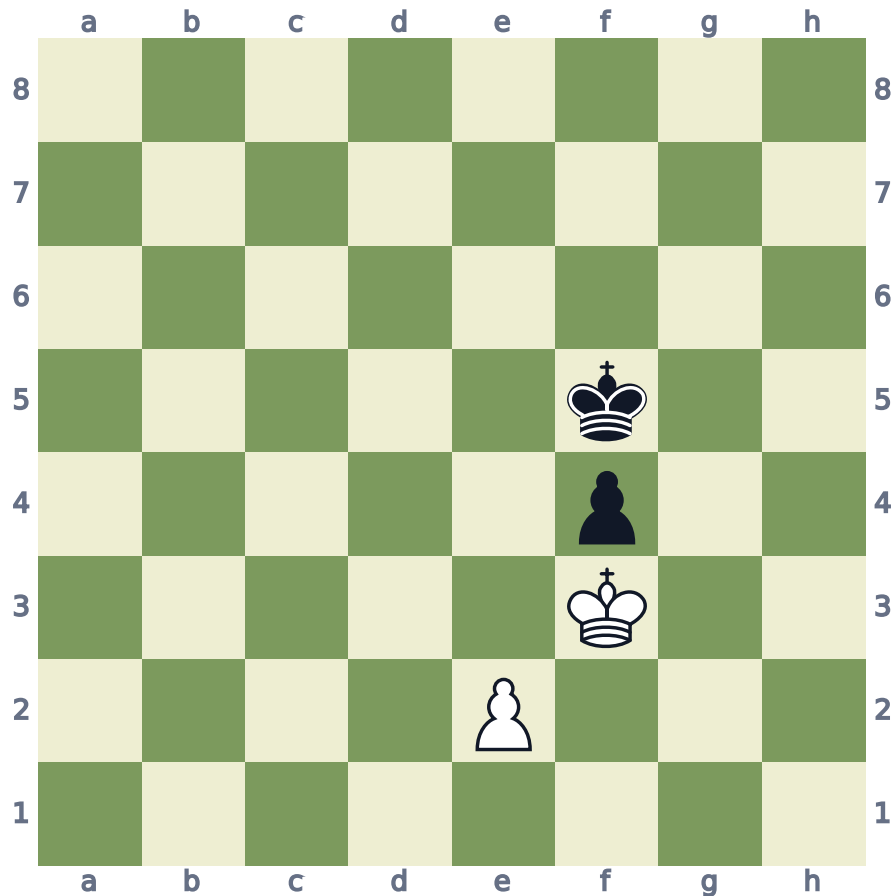
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Minor-piece endings contain tactical races. Calculate pawn breakthroughs, sacrifices, and zugzwangs before applying a general rule.

BOARD LABORATORY 20

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Sacrifice a Minor Piece for the Last Pawns When It Draws'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Material value depends on winning capacity. Eliminating every pawn may create an immediate theoretical draw despite losing the piece. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Set up eight endings about sacrifice a minor piece for the last pawns when it draws. Play each against a training partner or engine from both sides and record the decisive route.

LEARNING UNIT 21

Cut Off the Enemy King With the Rook

THE RULE

Use a rook along a rank or file to deny the king access before pursuing pawns.

WHY IT WORKS

King restriction gains virtual tempi. Your king can improve while the opposing king remains separated from the critical area.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

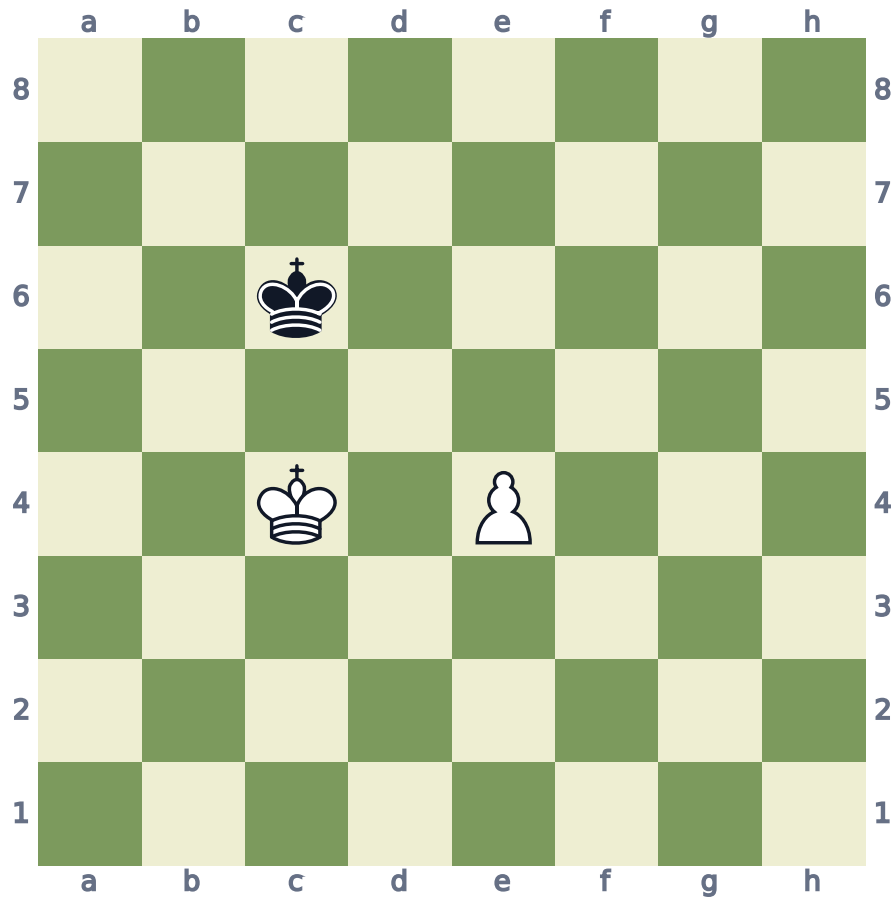
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 21

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Cut Off the Enemy King With the Rook'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: King restriction gains virtual tempi. Your king can improve while the opposing king remains separated from the critical area. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving cut off the enemy king with the rook. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 22

Build the Lucena Bridge

THE RULE

In the standard rook-and-pawn win, use rook checks and shielding to escort the king from the promotion file.

WHY IT WORKS

The bridge blocks side checks and lets the pawn promote. Exact rook placement converts an advanced pawn into a technical win.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

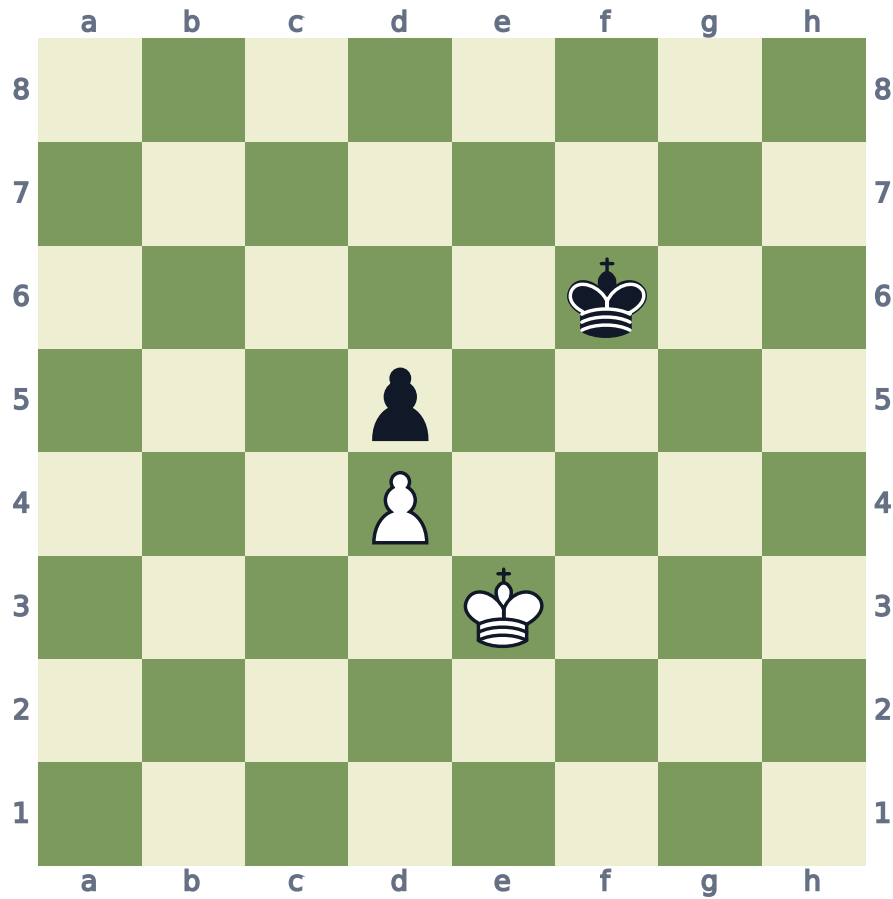
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 22

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Build the Lucena Bridge'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The bridge blocks side checks and lets the pawn promote. Exact rook placement converts an advanced pawn into a technical win. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving build the lucena bridge. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 23

Use the Philidor Third-Rank Defense

THE RULE

Keep the defending rook active on the third rank until the stronger king advances, then check from behind.

WHY IT WORKS

The setup prevents king shelter and preserves checking distance. Timing the retreat avoids passive front-blockade failure.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

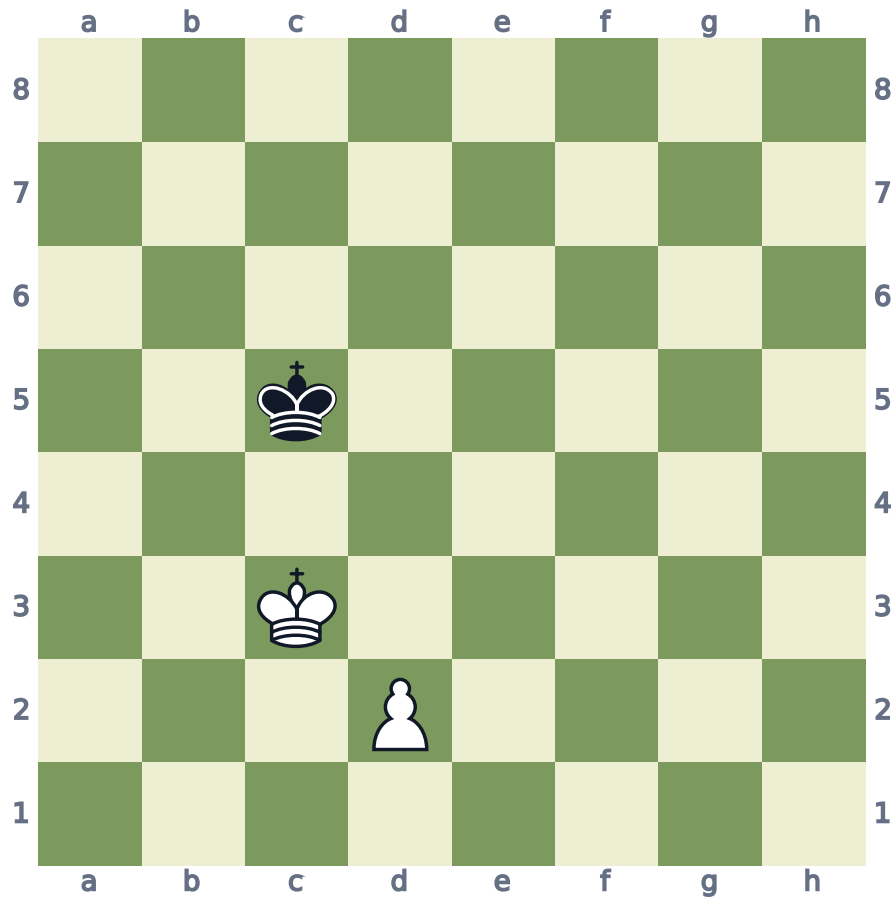
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 23

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Use the Philidor Third-Rank Defense'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The setup prevents king shelter and preserves checking distance. Timing the retreat avoids passive front-blockade failure. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving use the philidor third-rank defense. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 24

Check From the Side With Adequate Distance

THE RULE

Use lateral rook checks when at least several files separate the rook from the king.

WHY IT WORKS

Distance prevents the king from attacking the rook while checks hold the king away from the pawn's support squares.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

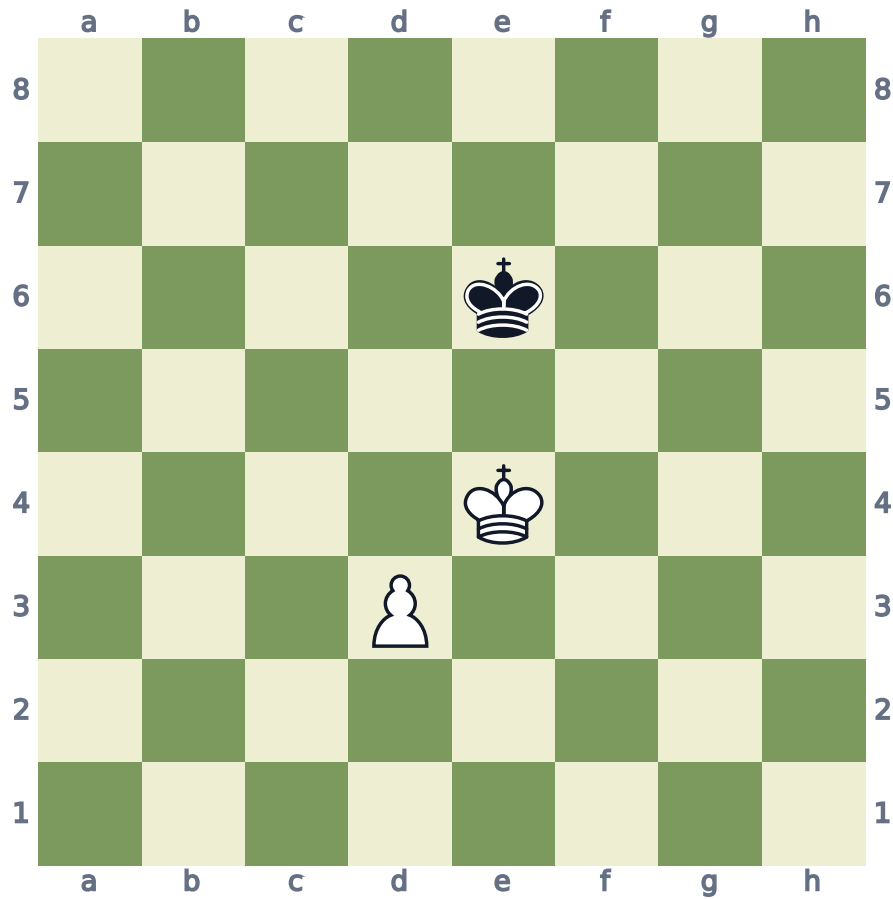
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 24

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Check From the Side With Adequate Distance'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Distance prevents the king from attacking the rook while checks hold the king away from the pawn's support squares. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving check from the side with adequate distance. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 25

Know the Wrong-Bishop Rook-Pawn Draw

THE RULE

Recognize that a bishop unable to control the pawn's promotion corner may not win with a rook pawn.

WHY IT WORKS

The defending king reaches the corner and cannot be displaced without stalemate. Extra material does not create the required square control.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

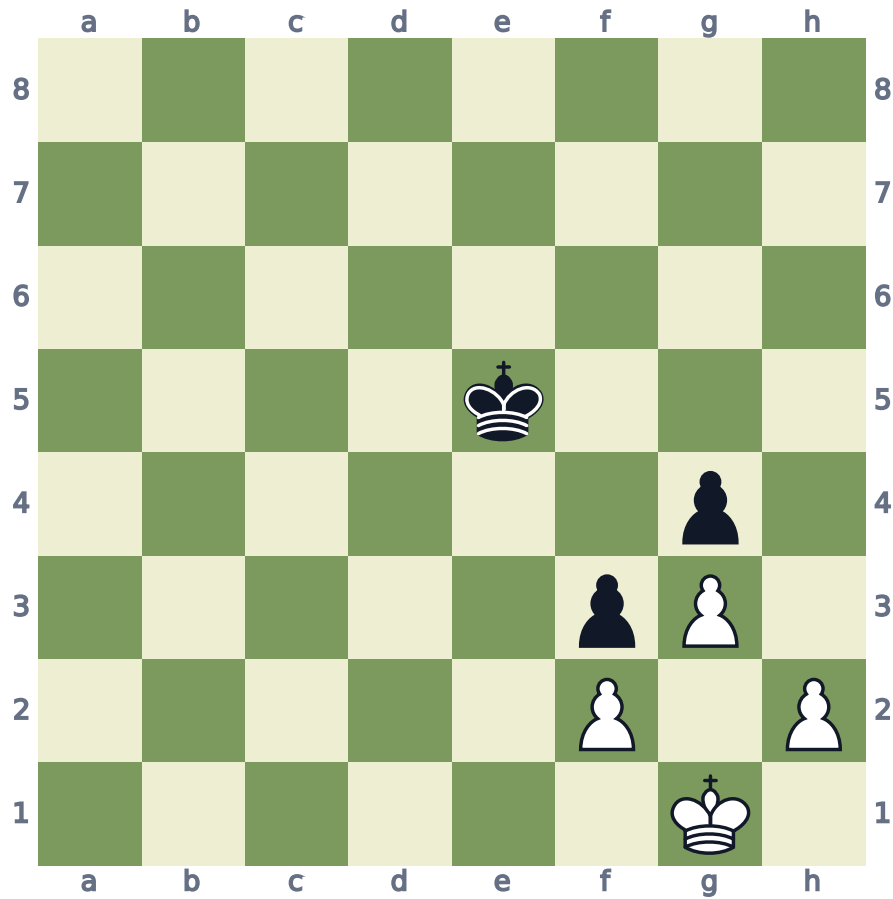
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 25

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Know the Wrong-Bishop Rook-Pawn Draw'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The defending king reaches the corner and cannot be displaced without stalemate. Extra material does not create the required square control. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving know the wrong-bishop rook-pawn draw. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 26

Centralize the Queen With Tempo

THE RULE

Use checks and attacks to bring the queen toward the center without allowing counterchecks.

WHY IT WORKS

A centralized queen controls many checking routes and can switch wings instantly, but exposed kings make move order decisive.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

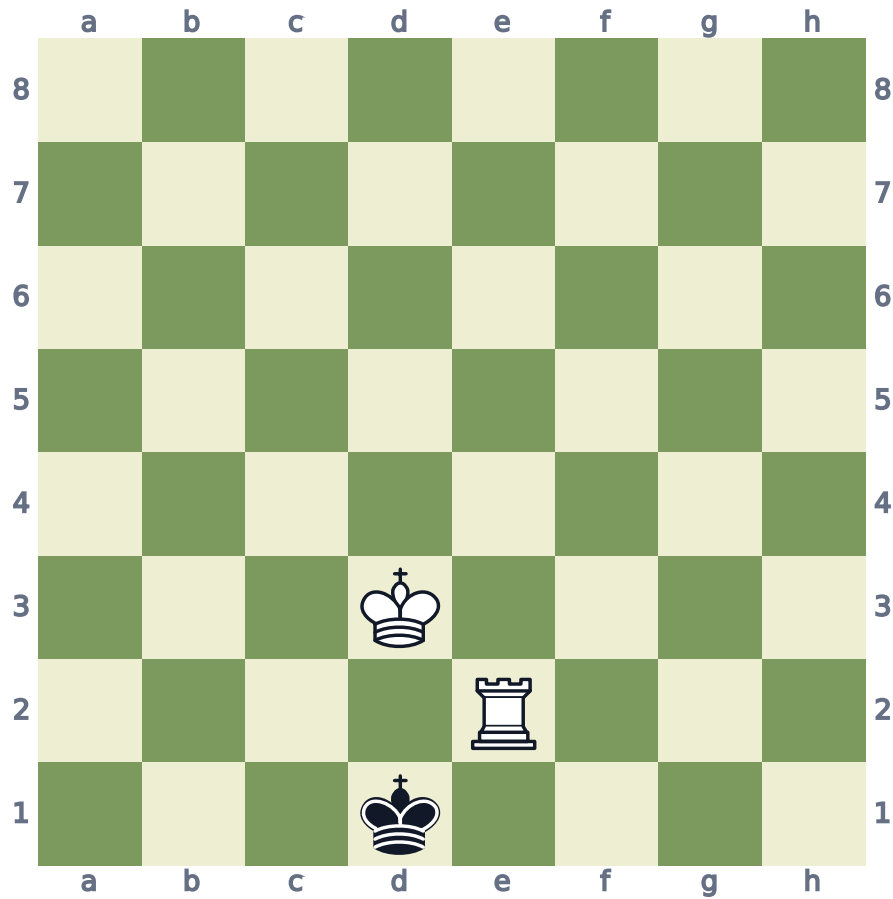
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 26

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Centralize the Queen With Tempo'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A centralized queen controls many checking routes and can switch wings instantly, but exposed kings make move order decisive. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving centralize the queen with tempo. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 27

King Safety Dominates Queen Endgames

THE RULE

Evaluate checking networks and perpetuals before counting pawns.

WHY IT WORKS

Queens create long forcing sequences. One exposed king can make several extra pawns irrelevant because conversion is impossible without escaping checks.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

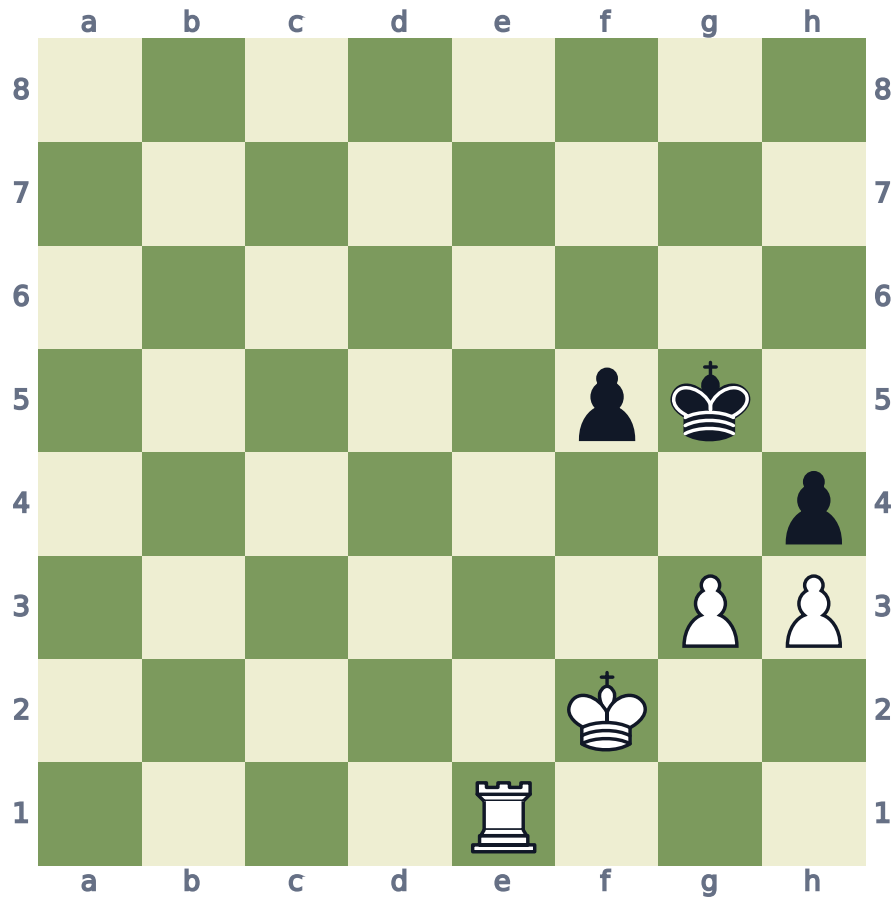
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 27

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'King Safety Dominates Queen Endgames'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Queens create long forcing sequences. One exposed king can make several extra pawns irrelevant because conversion is impossible without escaping checks. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving king safety dominates queen endgames. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 28

Build a Perpetual-Check Map

THE RULE

Identify a repeatable set of checking squares before beginning the sequence.

WHY IT WORKS

A sustainable perpetual requires the king to cycle without a safe shelter or queen trade. Mapping squares prevents running out of checks.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

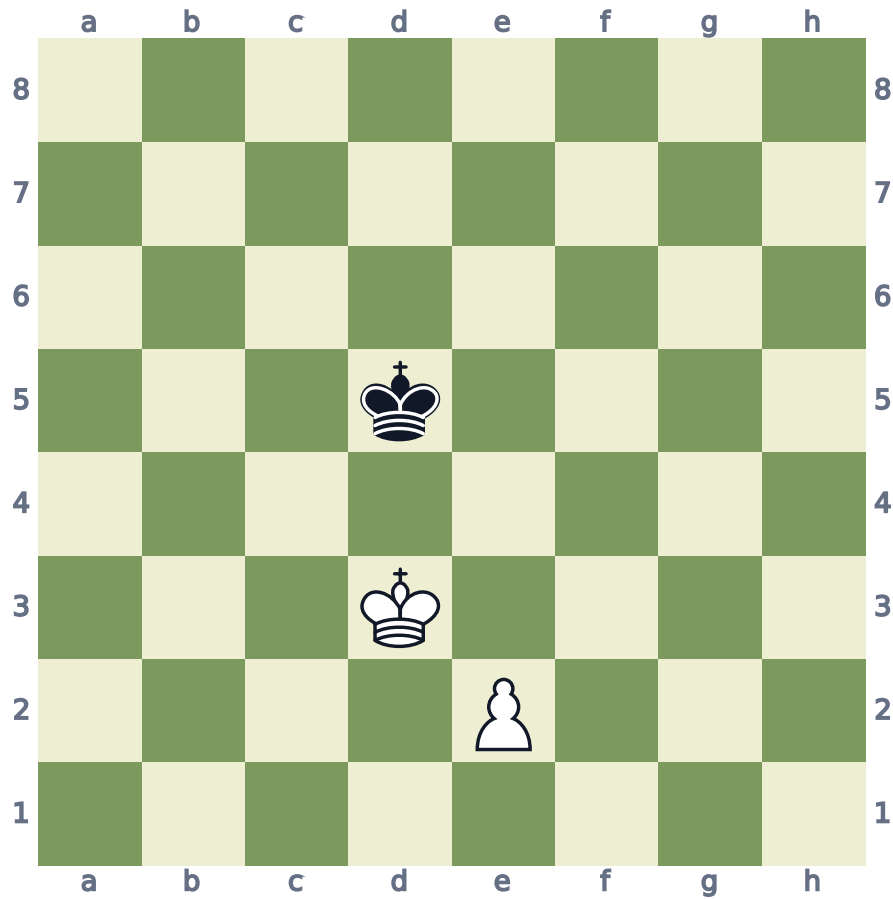
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 28

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Build a Perpetual-Check Map'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A sustainable perpetual requires the king to cycle without a safe shelter or queen trade. Mapping squares prevents running out of checks. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving build a perpetual-check map. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 29

Calculate Queen Trades to the Final Pawn Ending

THE RULE

Before offering or accepting a queen trade, calculate the resulting king-and-pawn position exactly.

WHY IT WORKS

Queen exchanges can instantly transform an active draw into a lost pawn ending or a dangerous win into a simple draw.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

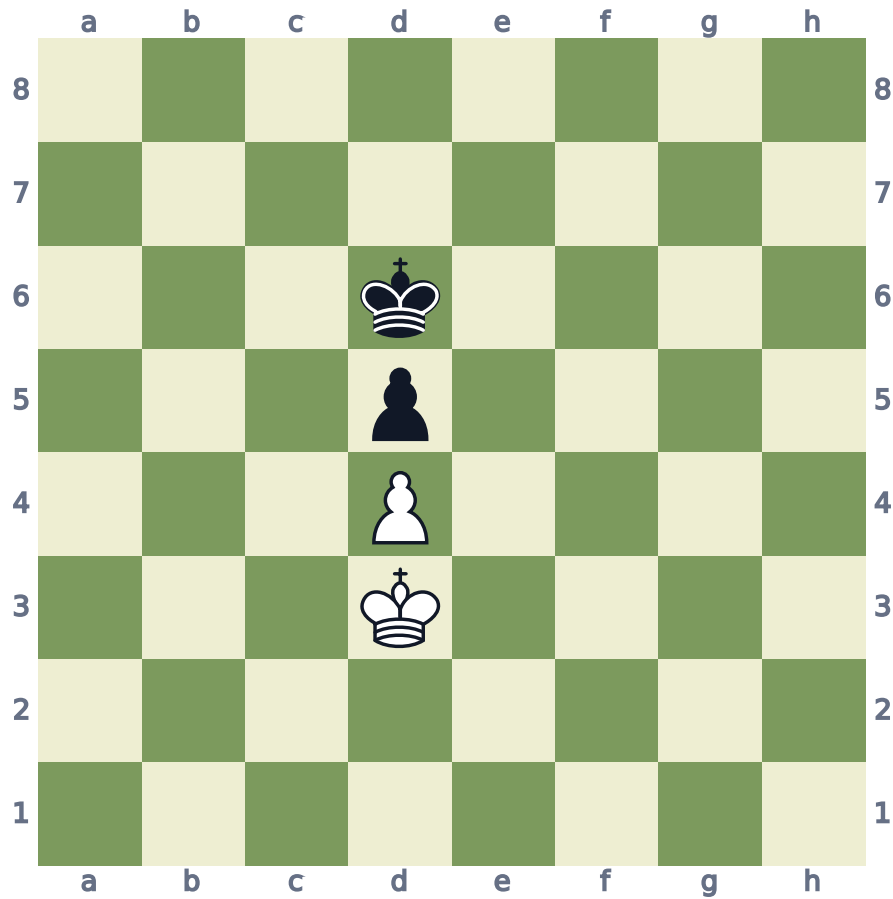
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 29

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Calculate Queen Trades to the Final Pawn Ending'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Queen exchanges can instantly transform an active draw into a lost pawn ending or a dangerous win into a simple draw. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving calculate queen trades to the final pawn ending. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 30

Connected Passed Pawns Can Outweigh a Rook

THE RULE

Coordinate connected advanced passers so the rook cannot stop both without help.

WHY IT WORKS

Each pawn protects or advances while the other forces the rook's placement. Sufficiently advanced connected pawns can dominate a lone rook.

RECOGNITION CUES

Heavy pieces remain with exposed kings or passed pawns; checking distance and activity matter; a technical position is approaching

APPLY IT OVER THE BOARD

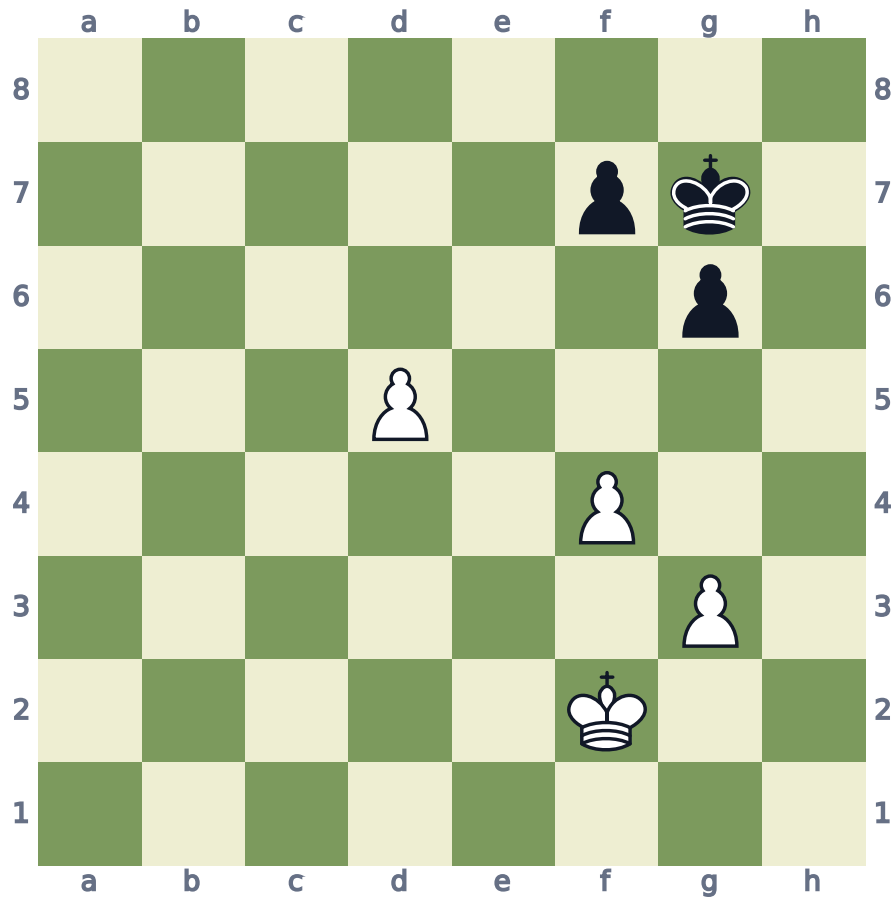
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Heavy-piece endings are check-sensitive. A single tempo, checking angle, or king exposure can outweigh a pawn or textbook placement.

BOARD LABORATORY 30

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Connected Passed Pawns Can Outweigh a Rook'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Each pawn protects or advances while the other forces the rook's placement. Sufficiently advanced connected pawns can dominate a lone rook. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Practice ten technical positions involving connected passed pawns can outweigh a rook. Calculate checks first, then compare activity, pawn count, and king safety.

LEARNING UNIT 31

Win the Key Squares

THE RULE

In king-and-pawn endings, identify the squares from which the king can force promotion regardless of opposition.

WHY IT WORKS

Key squares translate the promotion problem into king access. If the attacking king reaches one, it can escort the pawn or outflank the defender; if not, opposition may hold the draw.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

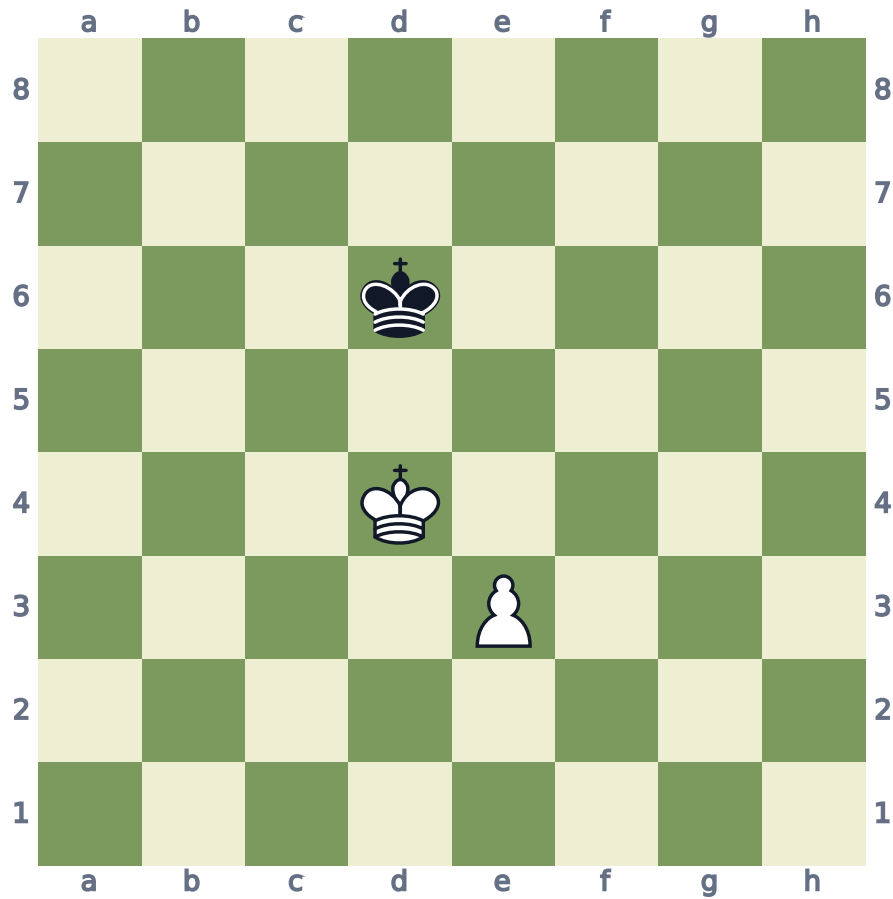
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 31

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Win the Key Squares'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Key squares translate the promotion problem into king access. If the attacking king reaches one, it can escort the pawn or outflank the defender; if not, opposition may hold the draw. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Win the Key Squares'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; score...

LEARNING UNIT 32

Use the Rule of the Square Precisely

THE RULE

Test whether a king can catch a passed pawn by entering its promotion square, then account for move order and interference.

WHY IT WORKS

The square gives a rapid geometric count without moving each piece. It is exact in the bare race, but checks, blocked routes, captures, and whose turn it is can alter the result.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

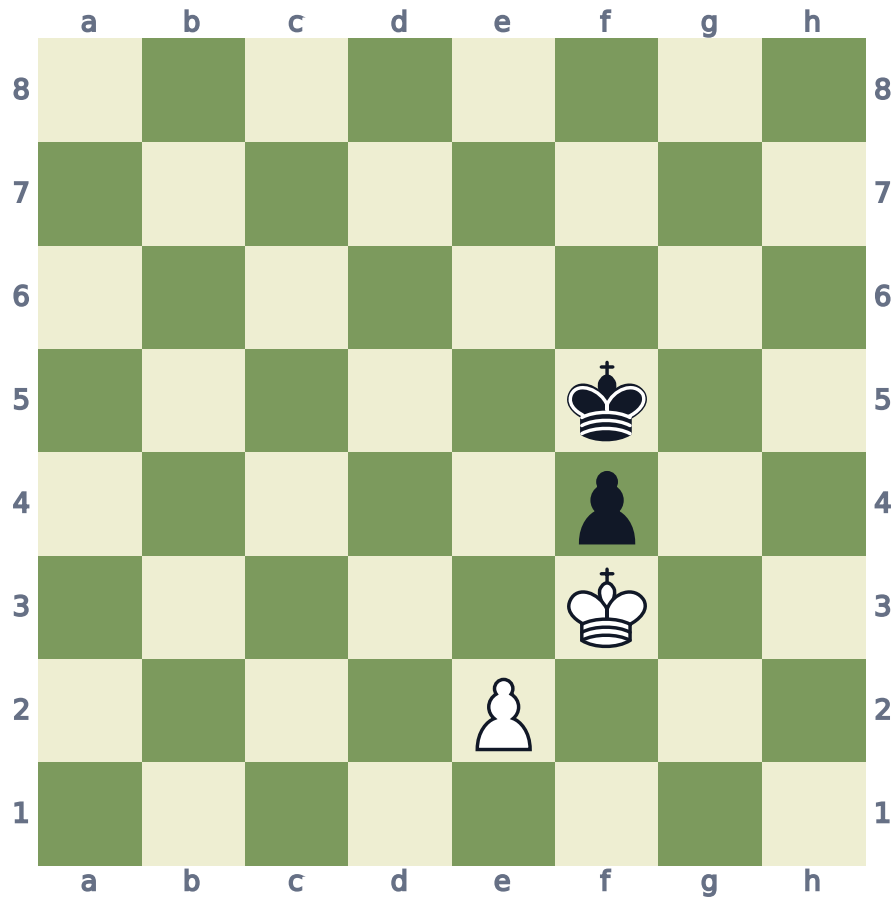
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 32

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Use the Rule of the Square Precisely'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The square gives a rapid geometric count without moving each piece. It is exact in the bare race, but checks, blocked routes, captures, and whose turn it is can alter the result. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Use the Rule of the Square Precisely'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 33

Map Corresponding Squares

THE RULE

In mutual zugzwang positions, identify pairs of squares where one king must answer the other's location.

WHY IT WORKS

Ordinary opposition is only the simplest correspondence. Complex pawn structures create networks of matching king squares; entering the network on the correct move forces the opponent to yield a critical route.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

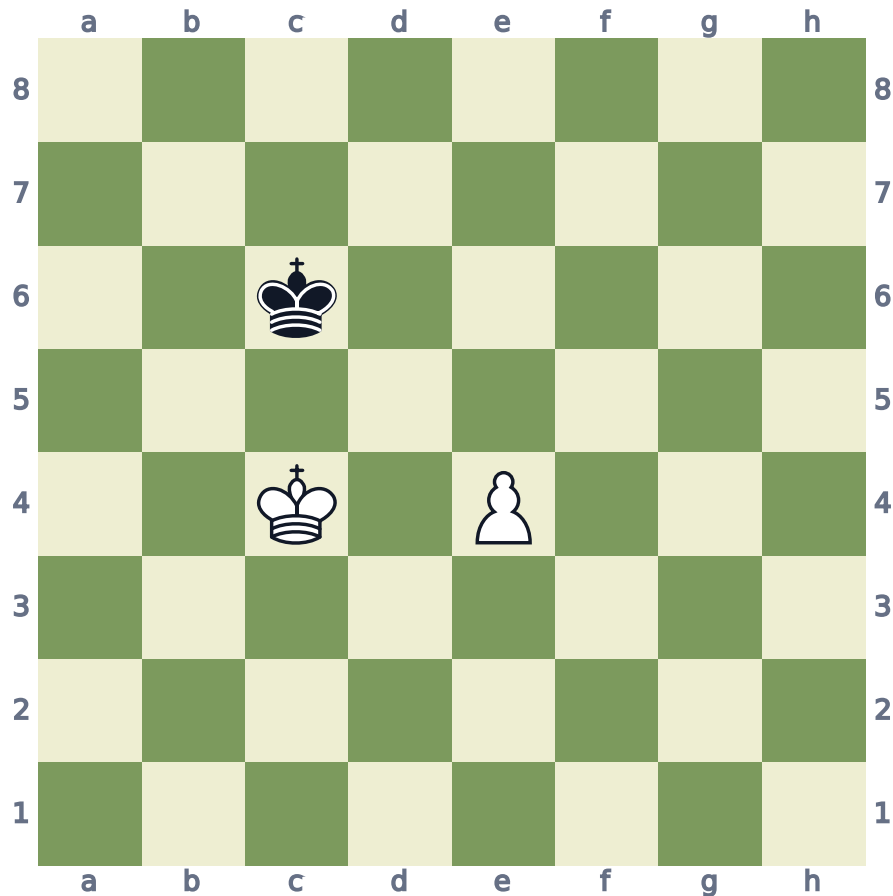
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 33

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Map Corresponding Squares'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Ordinary opposition is only the simplest correspondence. Complex pawn structures create networks of matching king squares; entering the network on the correct move forces the opponent to yield a critical route. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Map Corresponding Squares'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 34

Preserve a Reserve Tempo

THE RULE

Keep an unnecessary pawn move available when the position may become zugzwang.

WHY IT WORKS

A spare pawn step passes the move without changing the critical king geometry. Spending it too early can reverse the opposition and turn a win into a draw or a draw into a loss.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

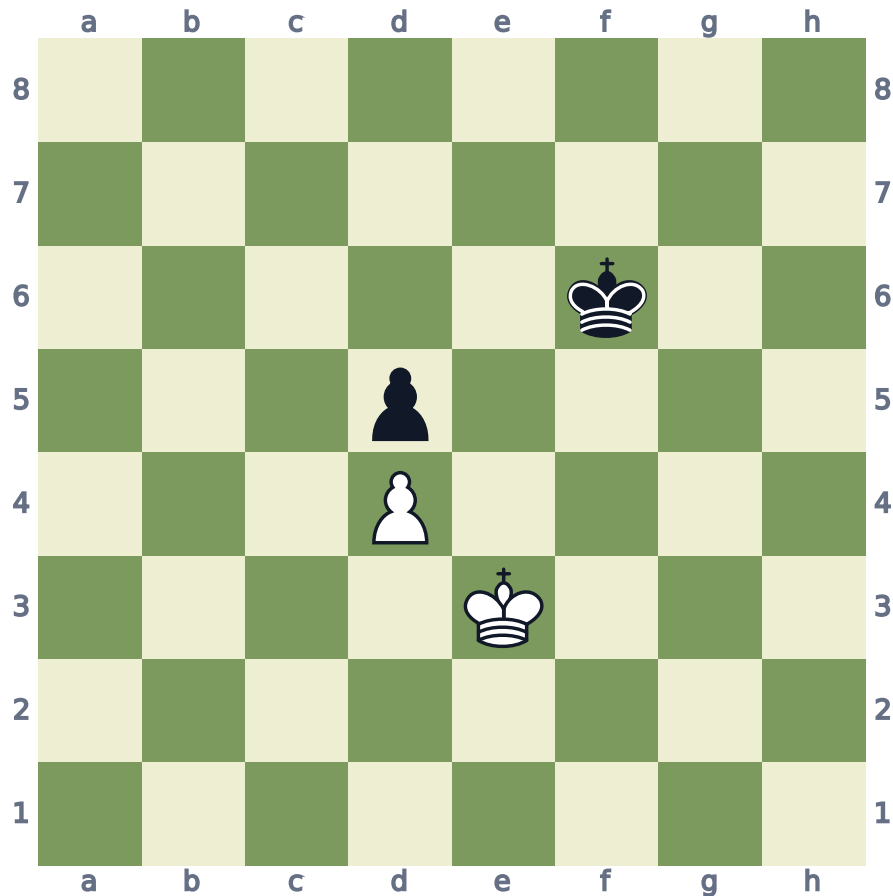
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 34

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Preserve a Reserve Tempo'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A spare pawn step passes the move without changing the critical king geometry. Spending it too early can reverse the opposition and turn a win into a draw or a draw into a loss. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Preserve a Reserve Tempo'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 35

Calculate the Pawn Breakthrough

THE RULE

Consider sacrificing one or two pawns when doing so forces a remaining pawn beyond the king's reach.

WHY IT WORKS

Connected pawns can divert enemy pawns by compulsory captures. The sacrifice changes files and tempos so that one survivor promotes before the king or remaining pawn can stop it.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

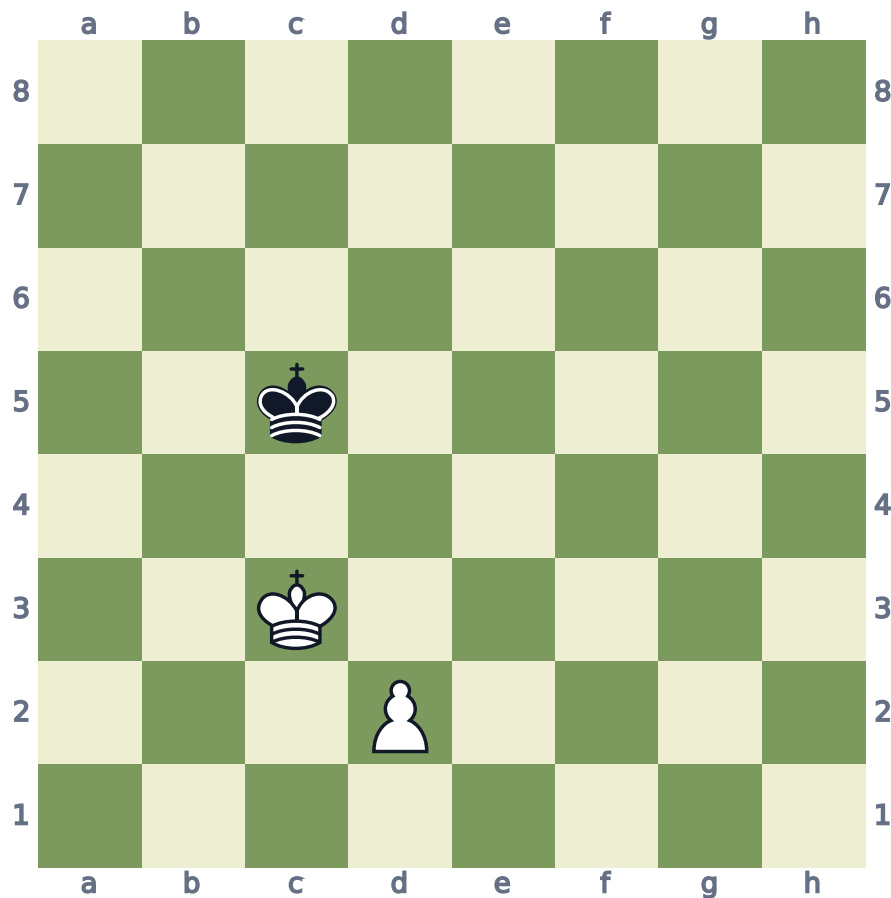
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 35

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Calculate the Pawn Breakthrough'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Connected pawns can divert enemy pawns by compulsory captures. The sacrifice changes files and tempos so that one survivor promotes before the king or remaining pawn can stop it. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Calculate the Pawn Breakthrough'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 36

Shoulder the Enemy King Away

THE RULE

Use your king's opposition and body to deny the rival king the shortest route.

WHY IT WORKS

Kings control adjacent squares and therefore act as moving barriers. A lateral approach can force the opposing king onto a longer path, winning the race even without immediately approaching the pawn.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

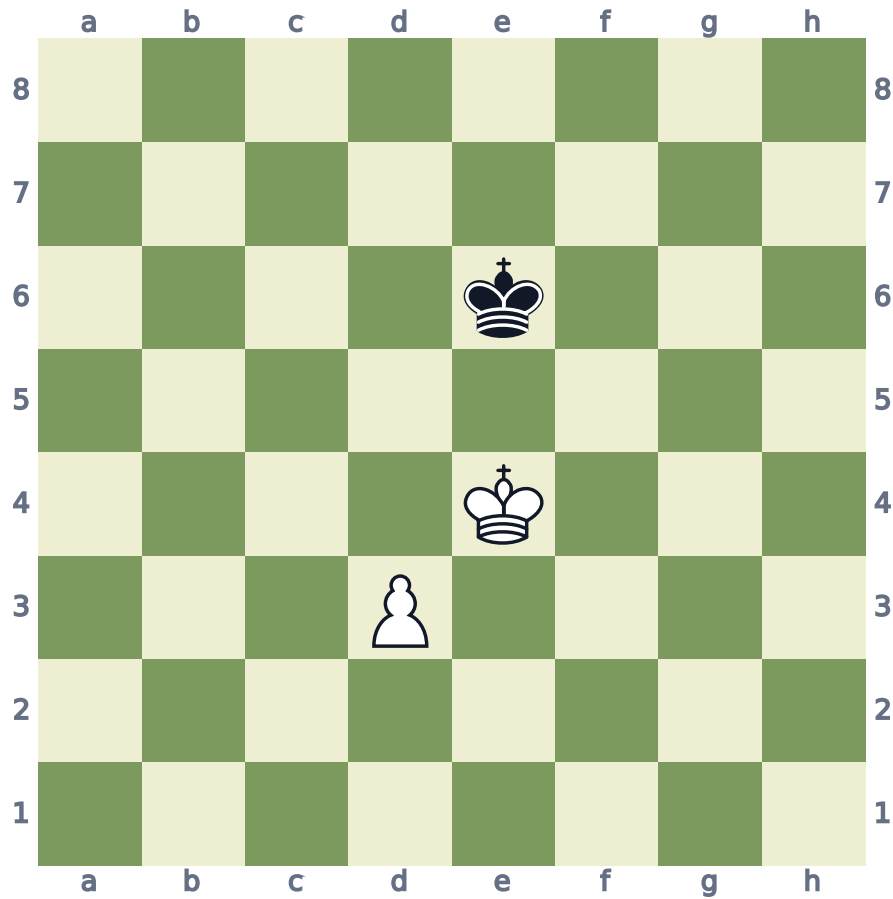
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 36

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Shoulder the Enemy King Away'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Kings control adjacent squares and therefore act as moving barriers. A lateral approach can force the opposing king onto a longer path, winning the race even without immediately approaching the pawn. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Shoulder the Enemy King Away'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 37

Use Distant Opposition

THE RULE

Recognize opposition even when three or five squares separate the kings on a file or rank.

WHY IT WORKS

With an odd number of squares between the kings, the player not to move often preserves the ability to take direct opposition later. This controls entry routes before the kings meet.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

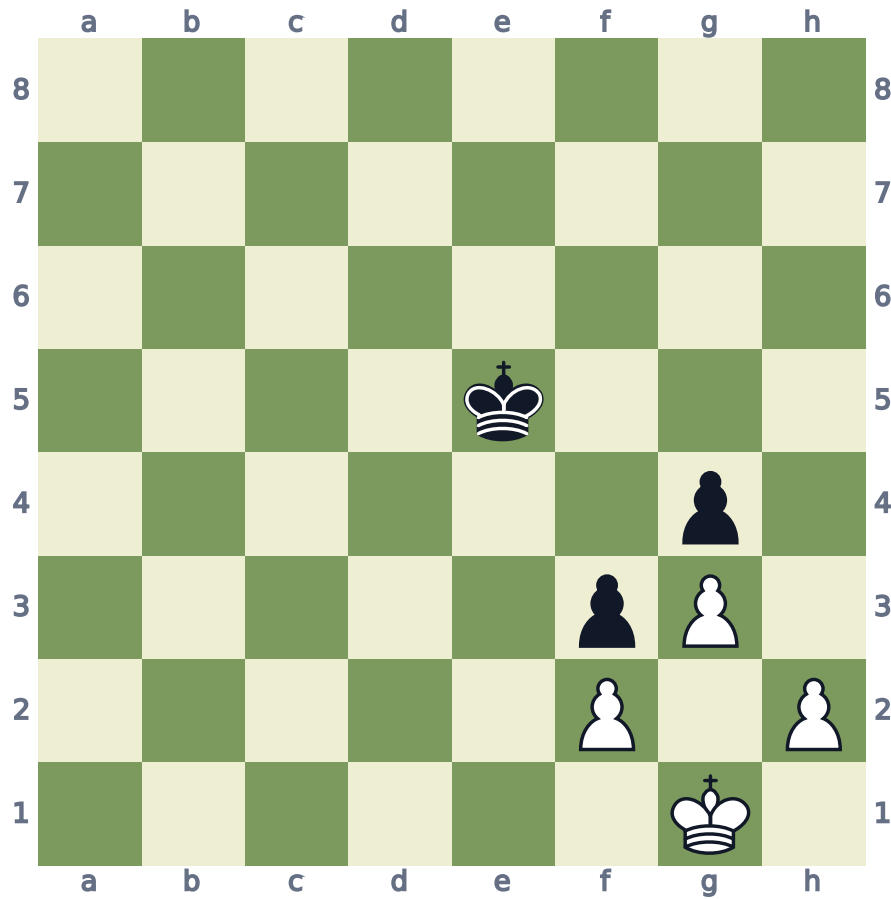
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 37

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Use Distant Opposition'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: With an odd number of squares between the kings, the player not to move often preserves the ability to take direct opposition later. This controls entry routes before the kings meet. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Use Distant Opposition'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; score...

LEARNING UNIT 38

Know the Knight's Rook-Pawn Problem

THE RULE

Do not assume a knight can stop or support a rook pawn when the key squares lie at the board's edge.

WHY IT WORKS

A knight loses half of its potential destinations near the rim and may be unable to both control promotion and avoid tempi. Exact geometry, especially the wrong-corner route, overrides material impressions.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

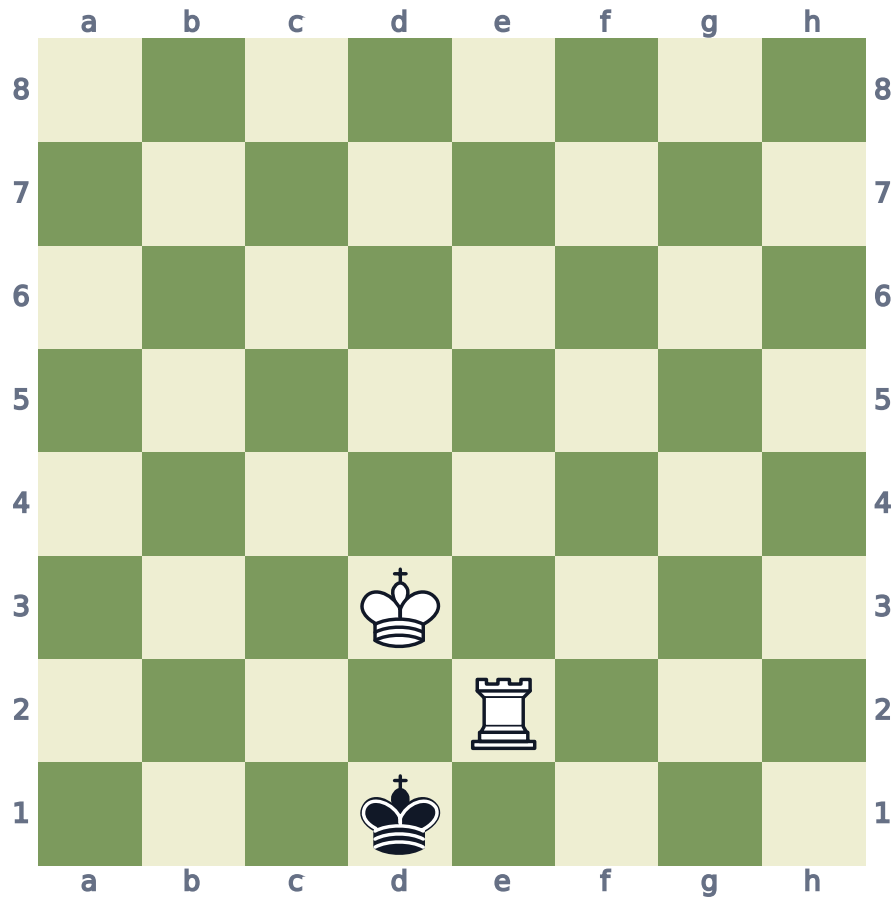
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 38

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Know the Knight's Rook-Pawn Problem'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A knight loses half of its potential destinations near the rim and may be unable to both control promotion and avoid tempi. Exact geometry, especially the wrong-corner route, overrides material impressions. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Know the Knight's Rook-Pawn Problem'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 39

Check the Bishop's Promotion Color

THE RULE

With a rook pawn and bishop, compare the promotion square's color with the bishop's color.

WHY IT WORKS

If the defending king reaches the corner that the bishop cannot control, the extra bishop and pawn may not dislodge it. The entire ending can hinge on one color relationship.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

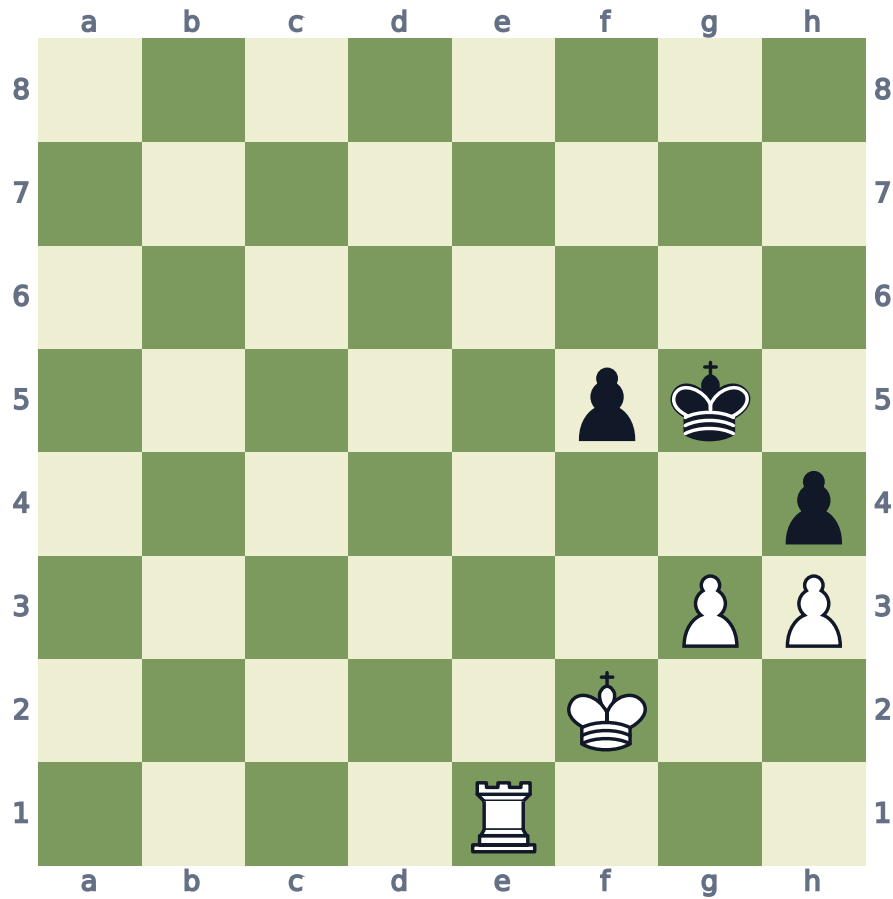
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 39

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Check the Bishop's Promotion Color'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: If the defending king reaches the corner that the bishop cannot control, the extra bishop and pawn may not dislodge it. The entire ending can hinge on one color relationship. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Check the Bishop's Promotion Color'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven...

LEARNING UNIT 40

Use the Outside Passer as a Decoy

THE RULE

Use a distant passer to pull the enemy king away from the main pawn mass.

WHY IT WORKS

The outside passer's value is diversion. Even when it is captured, the defending king may be drawn so far away that the attacking king wins the pawns on the other wing.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

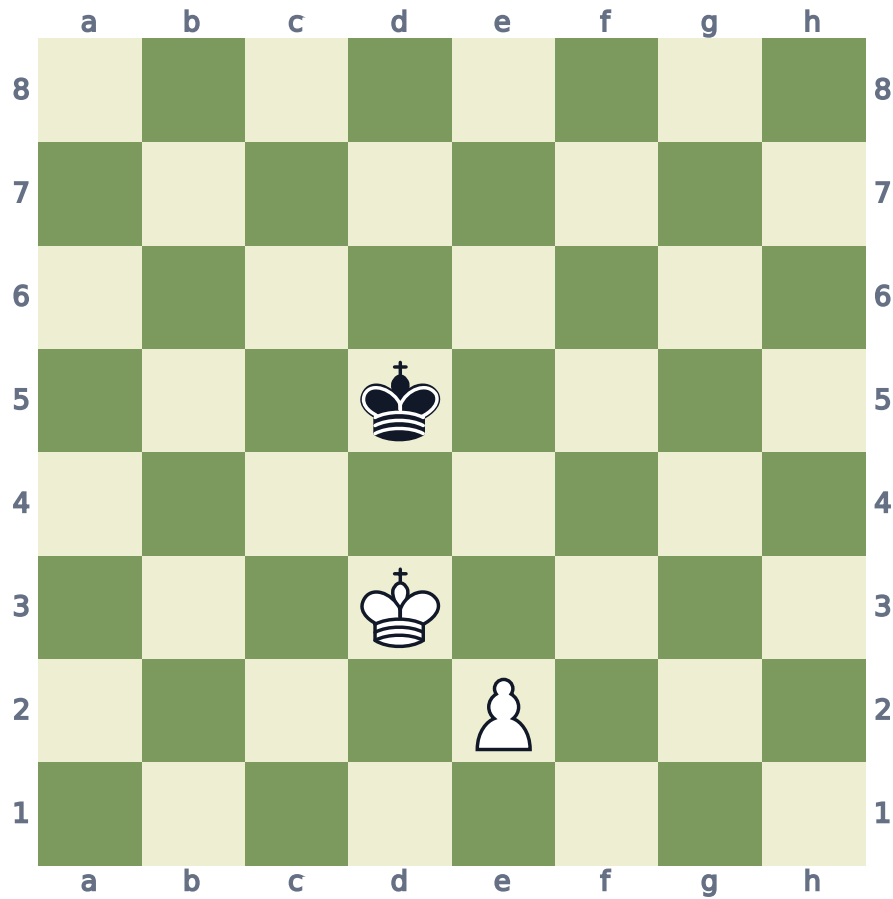
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 40

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Use the Outside Passer as a Decoy'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The outside passer's value is diversion. Even when it is captured, the defending king may be drawn so far away that the attacking king wins the pawns on the other wing. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Use the Outside Passer as a Decoy'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 41

Use a Protected Passer as a Barrier

THE RULE

A protected passed pawn can win by restricting the enemy king even before it advances.

WHY IT WORKS

The supporting pawn makes capture impossible, while the passer controls approach squares. The opposing king may be tied to the promotion zone, allowing your king to operate elsewhere.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

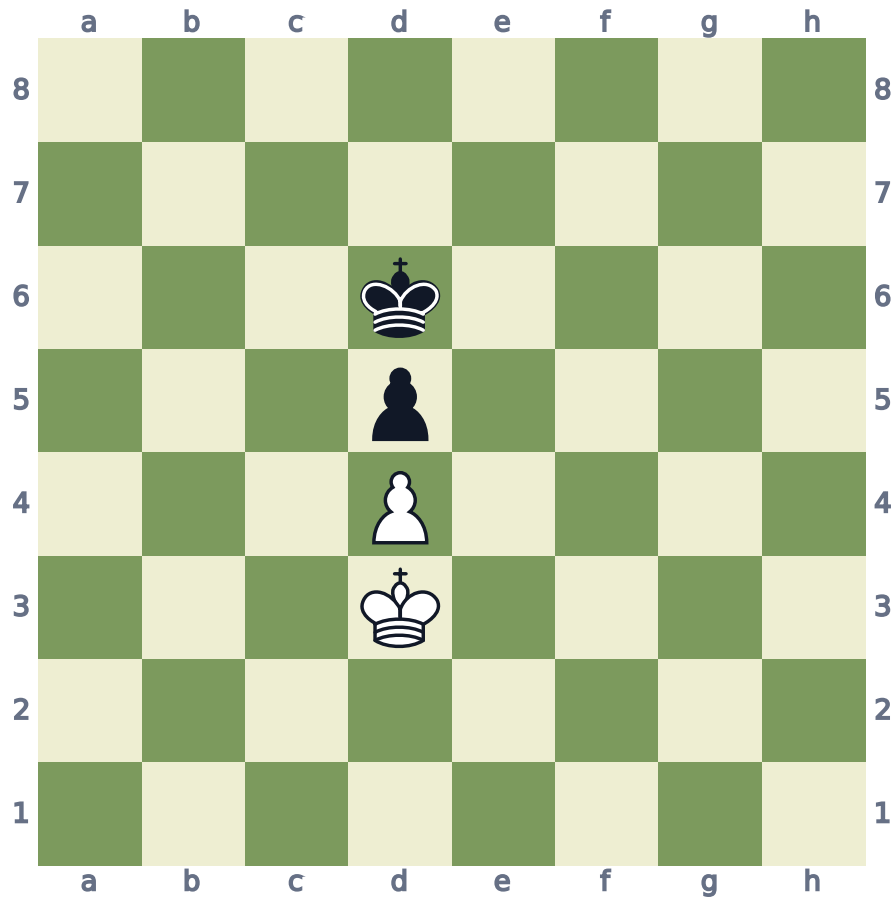
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 41

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Use a Protected Passer as a Barrier'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The supporting pawn makes capture impossible, while the passer controls approach squares. The opposing king may be tied to the promotion zone, allowing your king to operate elsewhere. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Use a Protected Passer as a Barrier'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 42

Respect Connected Passers on the Sixth

THE RULE

Two connected pawns on the sixth rank can often defeat a rook when they protect their advance correctly.

WHY IT WORKS

Advancing the rear pawn maintains mutual support and creates promotion threats on adjacent files. The rook may be unable to stop both without help from its king.

RECOGNITION CUES

few pieces remain; a king can approach a critical square; a pawn race or zugzwang is possible; one pawn move may spend the last reserve tempo

APPLY IT OVER THE BOARD

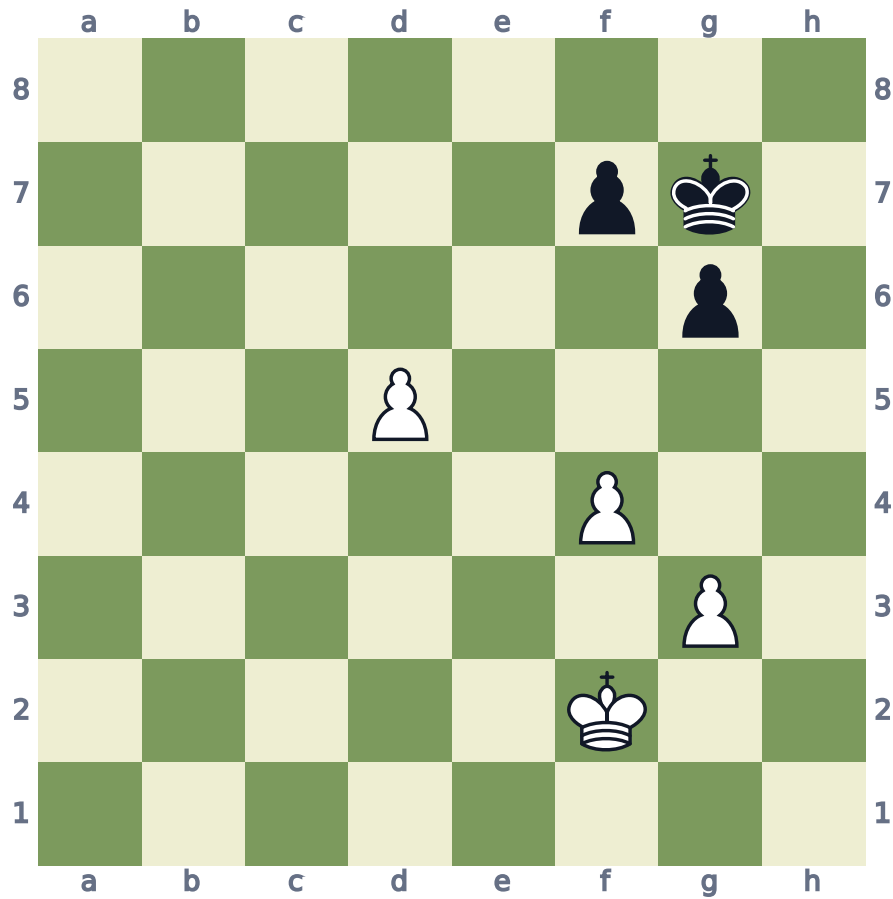
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Endgame rules are exact only under their stated conditions. Count every tempo, check for stalemate, and verify promotions, captures, and underpromotions on the board.

BOARD LABORATORY 42

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Respect Connected Passers on the Sixth'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Advancing the rear pawn maintains mutual support and creates promotion threats on adjacent files. The rook may be unable to stop both without help from its king. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Respect Connected Passers on the Sixth'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 43

Give Rook Checks from Maximum Distance

THE RULE

Keep checking distance so the enemy king cannot attack the rook while approaching.

WHY IT WORKS

A rook needs room to repeat checks. Distance increases the number of safe checking squares and forces the king to spend tempi finding shelter rather than advancing the pawn.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

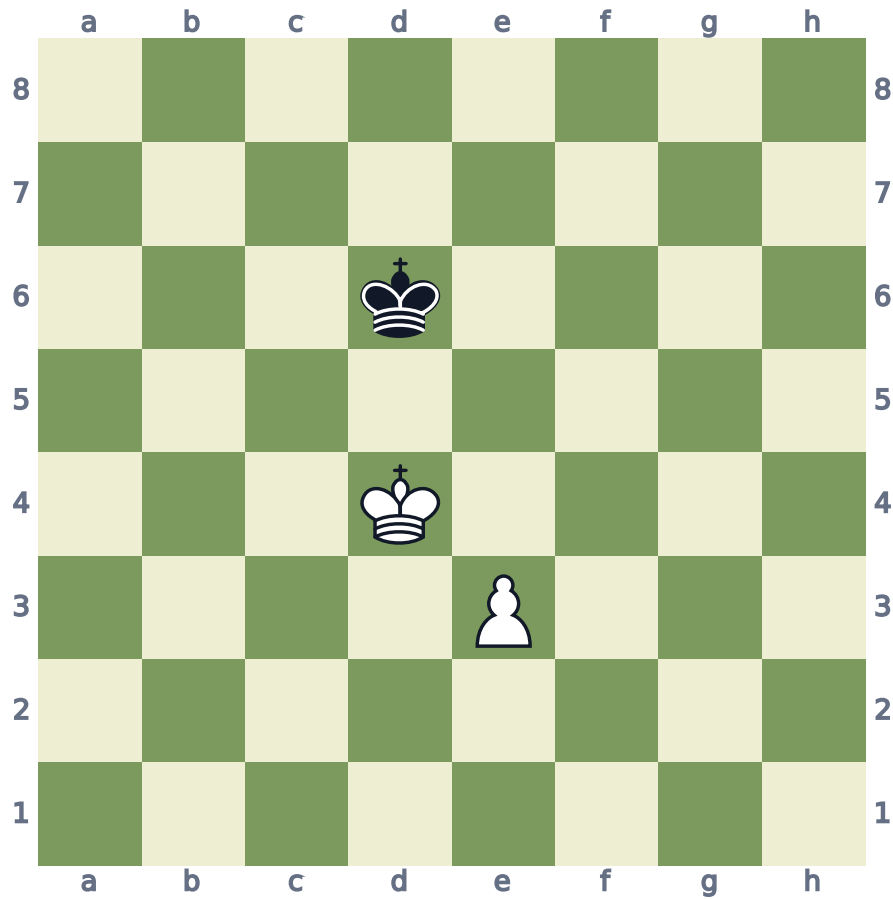
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 43

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Give Rook Checks from Maximum Distance'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A rook needs room to repeat checks. Distance increases the number of safe checking squares and forces the king to spend tempi finding shelter rather than advancing the pawn. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Give Rook Checks from Maximum Distance'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 44

Use the Seventh Rank as an Endgame Base

THE RULE

Place the rook on the opponent's second rank to attack pawns laterally and restrict the king.

WHY IT WORKS

A rook on the seventh combines multiple functions: it attacks pawn bases, gives side checks, and cuts the king. This activity can outweigh passively guarding one's own pawn.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

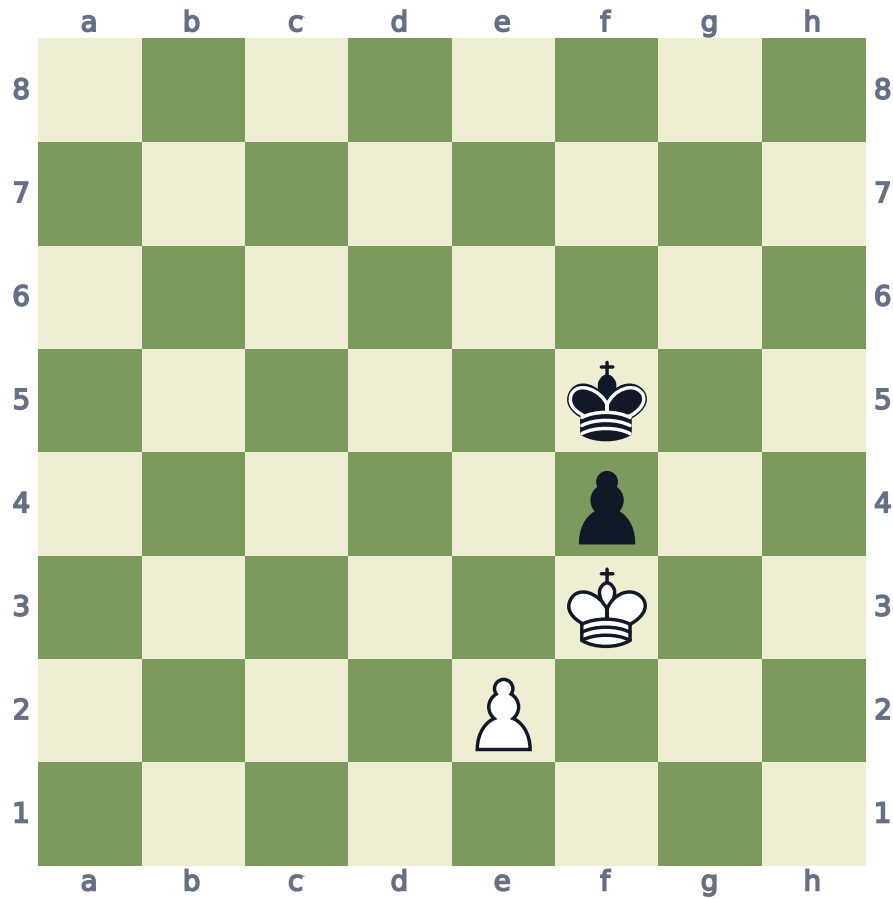
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 44

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Use the Seventh Rank as an Endgame Base'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A rook on the seventh combines multiple functions: it attacks pawn bases, gives side checks, and cuts the king. This activity can outweigh passively guarding one's own pawn. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Use the Seventh Rank as an Endgame Base'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 45

Learn the Vancura Defense

THE RULE

Against a rook pawn and rook, check the king from the side before it can shelter behind the pawn.

WHY IT WORKS

The Vancura setup draws by combining lateral checks with attacks on the pawn. Adequate side distance prevents the winning king from escaping checks while the defending king stays near promotion.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

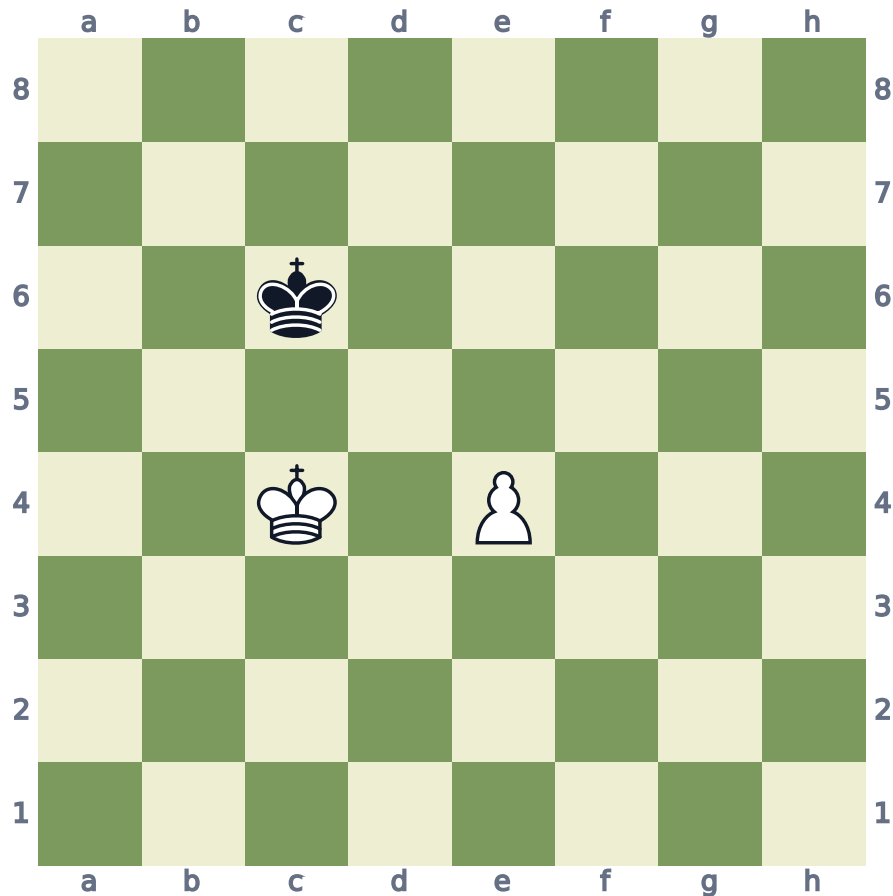
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 45

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Learn the Vancura Defense'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The Vancura setup draws by combining lateral checks with attacks on the pawn. Adequate side distance prevents the winning king from escaping checks while the defending king stays near promotion. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Learn the Vancura Defense'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 46

Defend from the Short Side

THE RULE

Place the defending king on the short side of the pawn so the rook retains long-side checking distance.

WHY IT WORKS

Separating the king and rook's tasks maximizes space for checks. If the rook is trapped on the short side, the attacking king can approach it and block the checking route.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

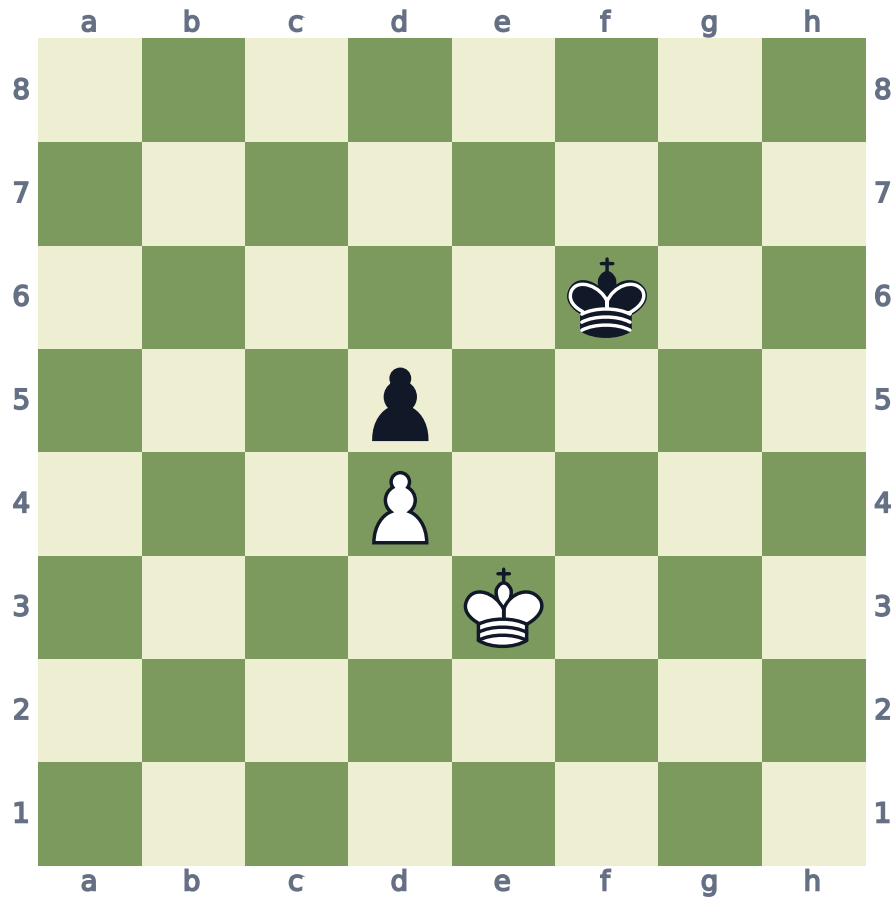
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 46

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Defend from the Short Side'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Separating the king and rook's tasks maximizes space for checks. If the rook is trapped on the short side, the attacking king can approach it and block the checking route. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Defend from the Short Side'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 47

Cut the King Off Before Pushing

THE RULE

Use the rook to exclude the enemy king by a file or rank before advancing the pawn.

WHY IT WORKS

Cutting off the king converts rook mobility into king-distance tempi. The defender must take a longer route, giving the attacking king time to escort the pawn or improve the rook.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

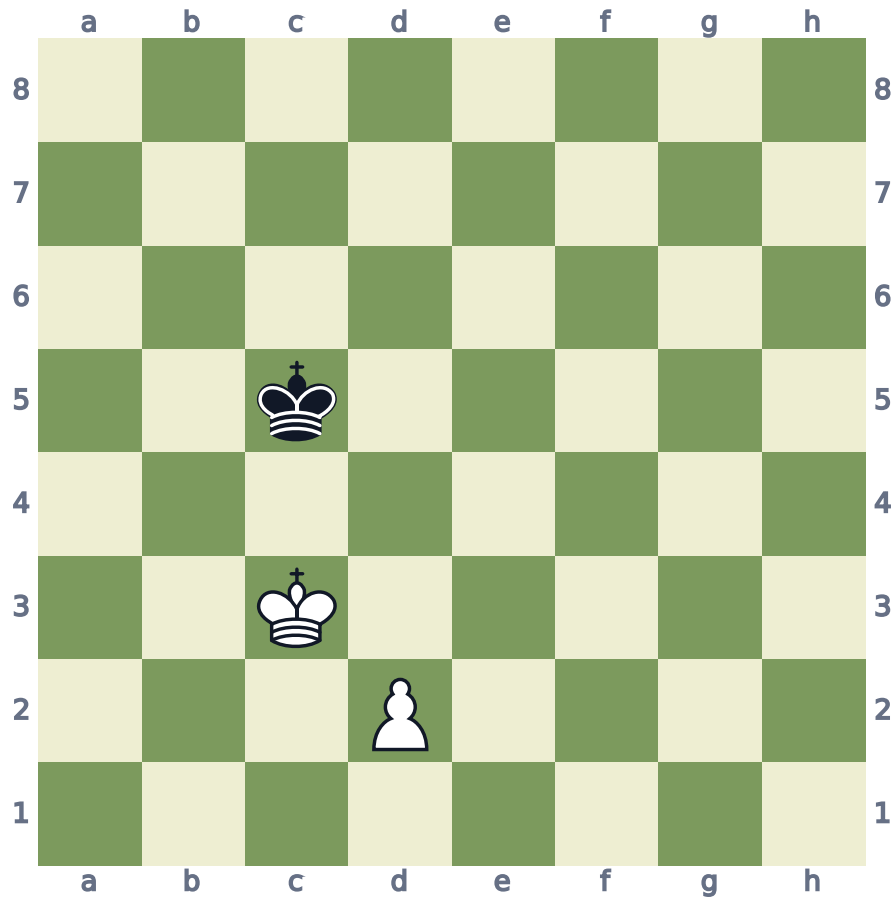
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 47

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Cut the King Off Before Pushing'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Cutting off the king converts rook mobility into king-distance tempi. The defender must take a longer route, giving the attacking king time to escort the pawn or improve the rook. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Cut the King Off Before Pushing'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 48

Verify the Pawn Ending Before Trading Rooks

THE RULE

Calculate the resulting king-and-pawn ending to its conclusion before exchanging rooks.

WHY IT WORKS

A rook advantage or active position can disappear into opposition, a pawn race, or a drawn rook-pawn ending. Exact liquidation replaces a complex advantage with a known result only when the count is correct.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

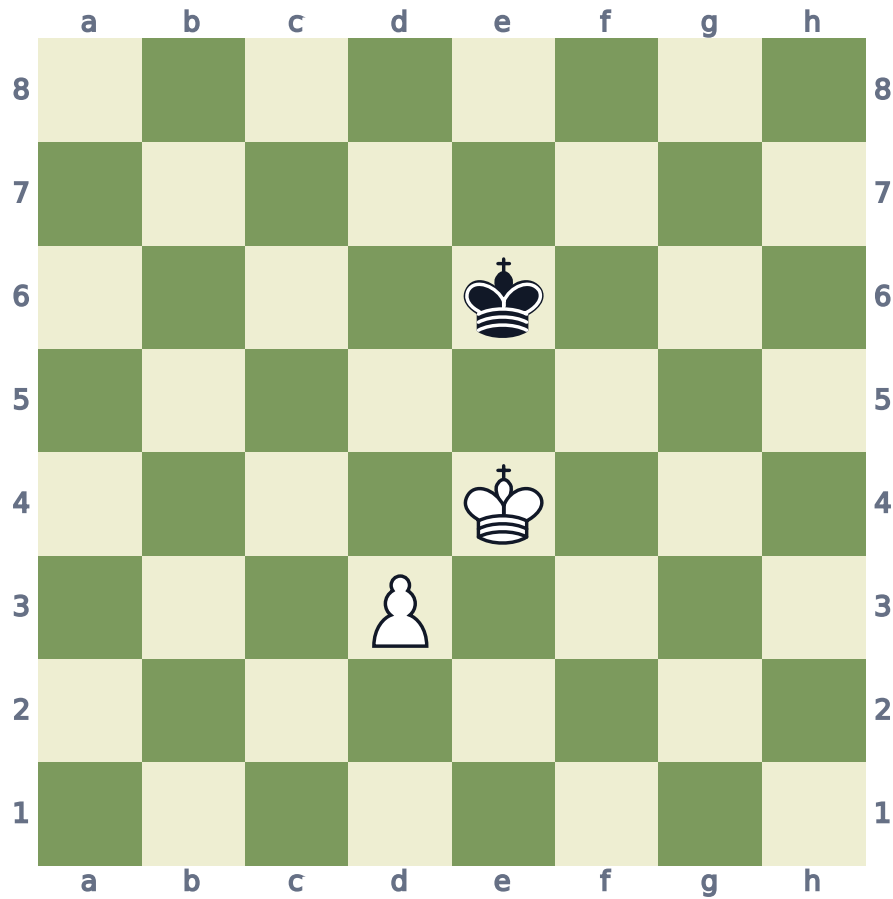
- Count material and passed pawns.
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- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 48

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Verify the Pawn Ending Before Trading Rooks'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A rook advantage or active position can disappear into opposition, a pawn race, or a drawn rook-pawn ending. Exact liquidation replaces a complex advantage with a known result only when the count is correct. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Verify the Pawn Ending Before Trading Rooks'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 49

Centralize the Queen with Check

THE RULE

In queen endings, improve the queen through checks that also approach central control.

WHY IT WORKS

A centralized queen reaches both wings, supports passed pawns, and offers more checking angles. Doing it with tempo prevents the opposing queen from establishing its own perpetual network.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

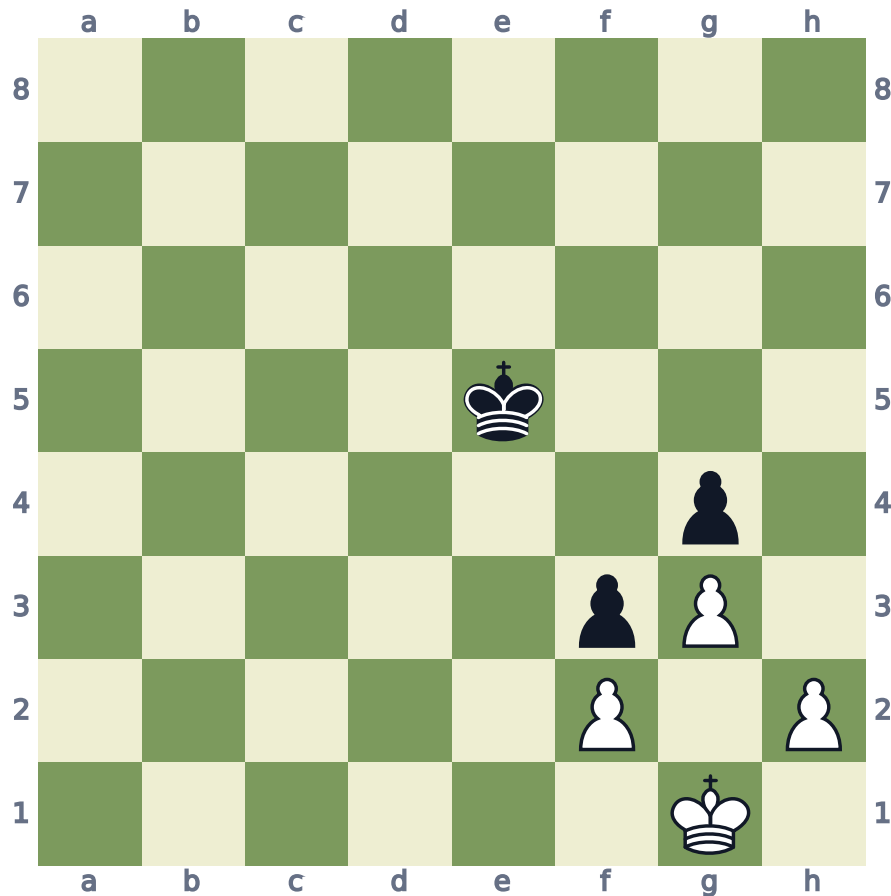
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 49

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Centralize the Queen with Check'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A centralized queen reaches both wings, supports passed pawns, and offers more checking angles. Doing it with tempo prevents the opposing queen from establishing its own perpetual network. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Centralize the Queen with Check'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 50

Control the Promotion Square in Queen versus Pawn

THE RULE

Use checks to force the king in front of its pawn, then approach with your king while controlling promotion.

WHY IT WORKS

The queen alone often cannot both check forever and capture safely. Forcing the king to block its pawn buys a tempo for the winning king to approach and ends stalemate tricks.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

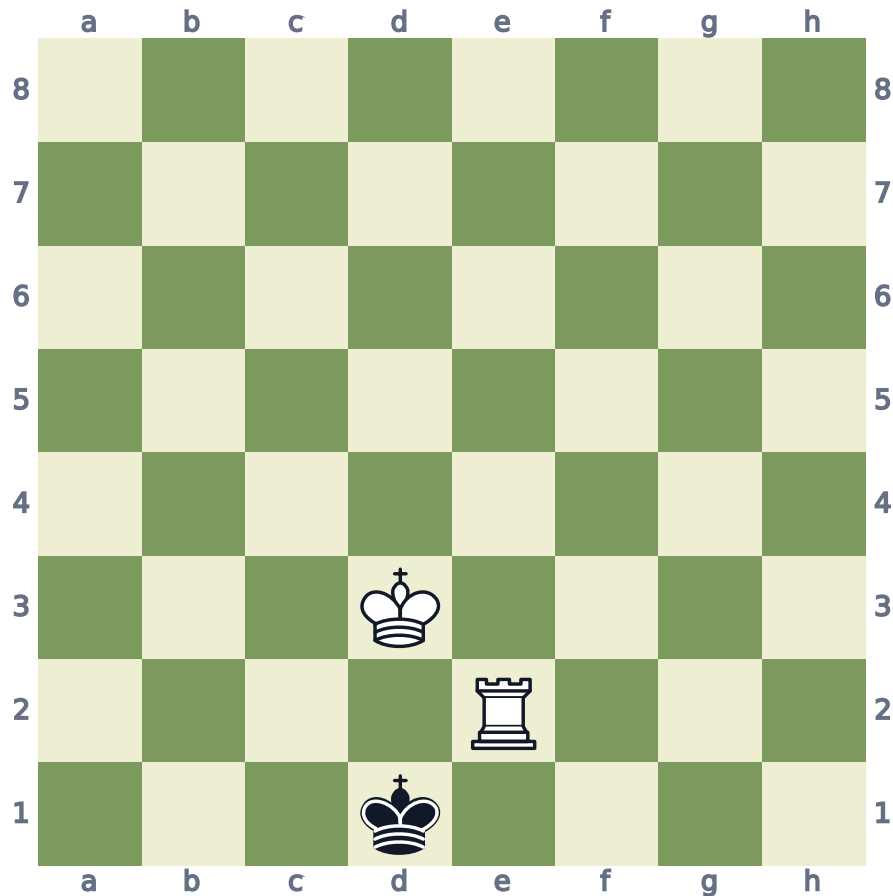
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 50

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Control the Promotion Square in Queen versus Pawn'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The queen alone often cannot both check forever and capture safely. Forcing the king to block its pawn buys a tempo for the winning king to approach and ends stalemate tricks. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Control the Promotion Square in Queen versus Pawn'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after...

LEARNING UNIT 51

Identify Perpetual-Check Zones

THE RULE

Before pushing a passed pawn, map the checks that could trap your king in a repeating zone.

WHY IT WORKS

Queen endings are governed by king shelter and checking geometry. A materially winning position can draw if the king cannot cross a rank or escape a sequence of diagonal and file checks.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

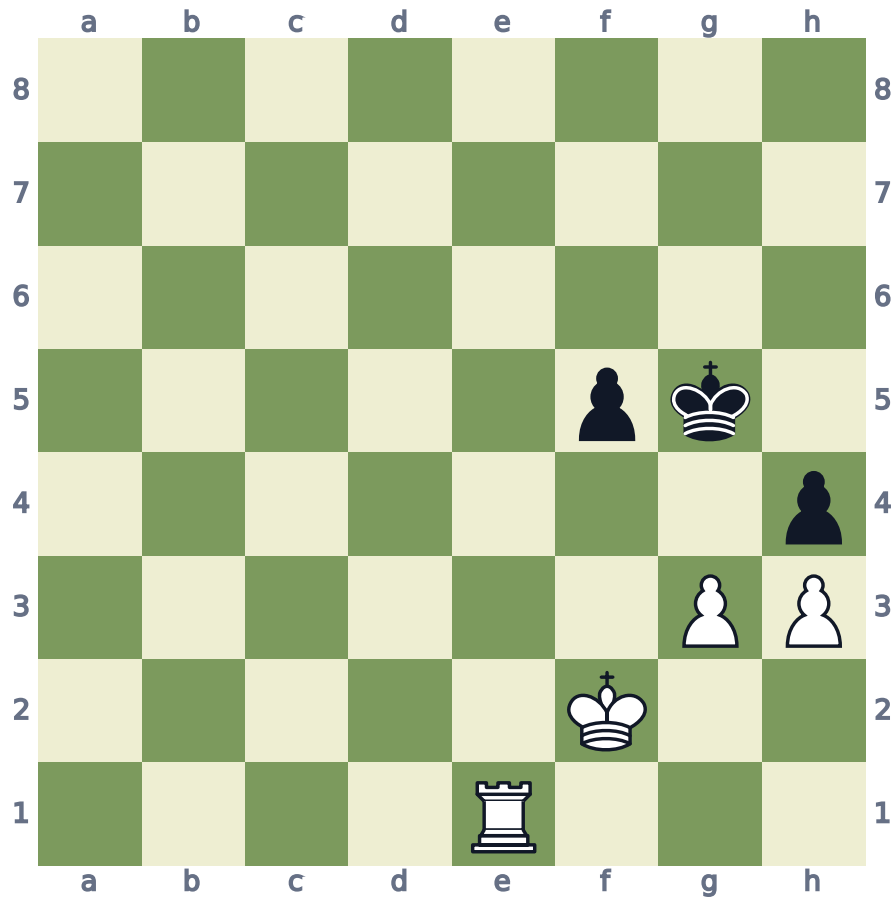
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 51

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Identify Perpetual-Check Zones'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Queen endings are governed by king shelter and checking geometry. A materially winning position can draw if the king cannot cross a rank or escape a sequence of diagonal and file checks. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Identify Perpetual-Check Zones'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 52

Trade Queens with a Cross-Check

THE RULE

Look for a checking move that interposes, attacks the enemy queen, or forces a queen exchange.

WHY IT WORKS

A cross-check changes the initiative because the opponent must answer your check before continuing theirs. This can break a perpetual and convert a queen ending into a winning pawn ending.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

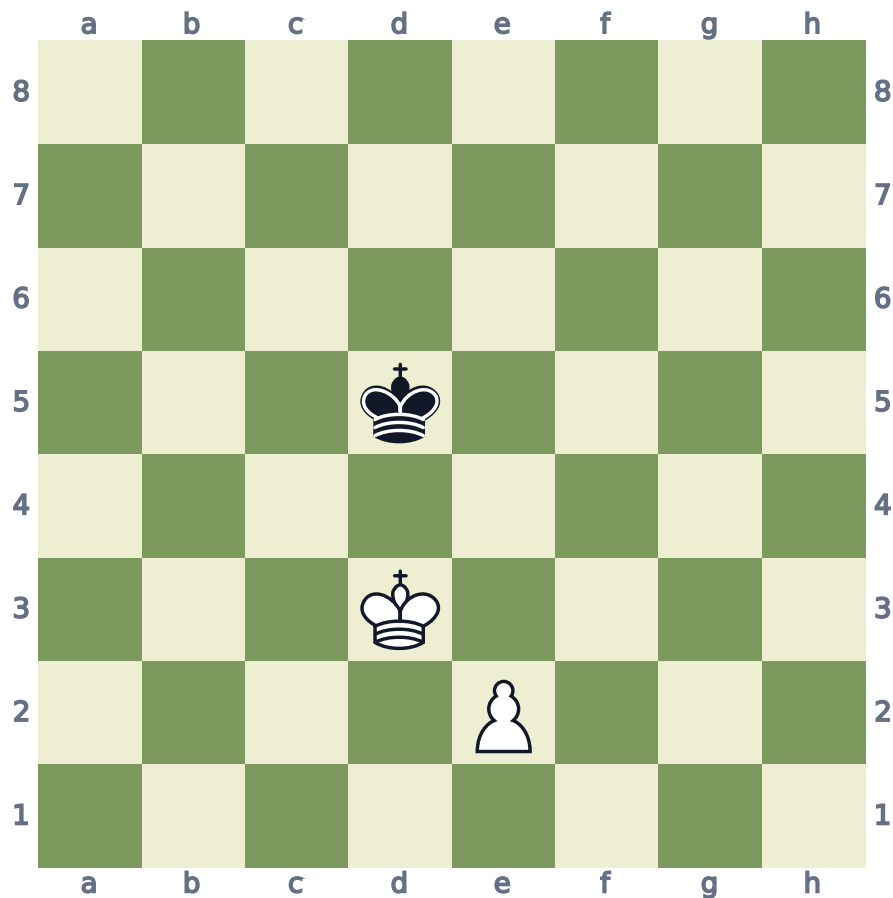
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 52

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Trade Queens with a Cross-Check'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A cross-check changes the initiative because the opponent must answer your check before continuing theirs. This can break a perpetual and convert a queen ending into a winning pawn ending. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Trade Queens with a Cross-Check'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 53

Build the Fortress Around One Entry Point

THE RULE

When materially worse, organize every piece around denying the opponent's only route of entry.

WHY IT WORKS

A fortress draws because the extra material cannot create a zugzwang, pawn break, or penetration. Identifying the single critical square allows a compact defense that never has to chase the attacker.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

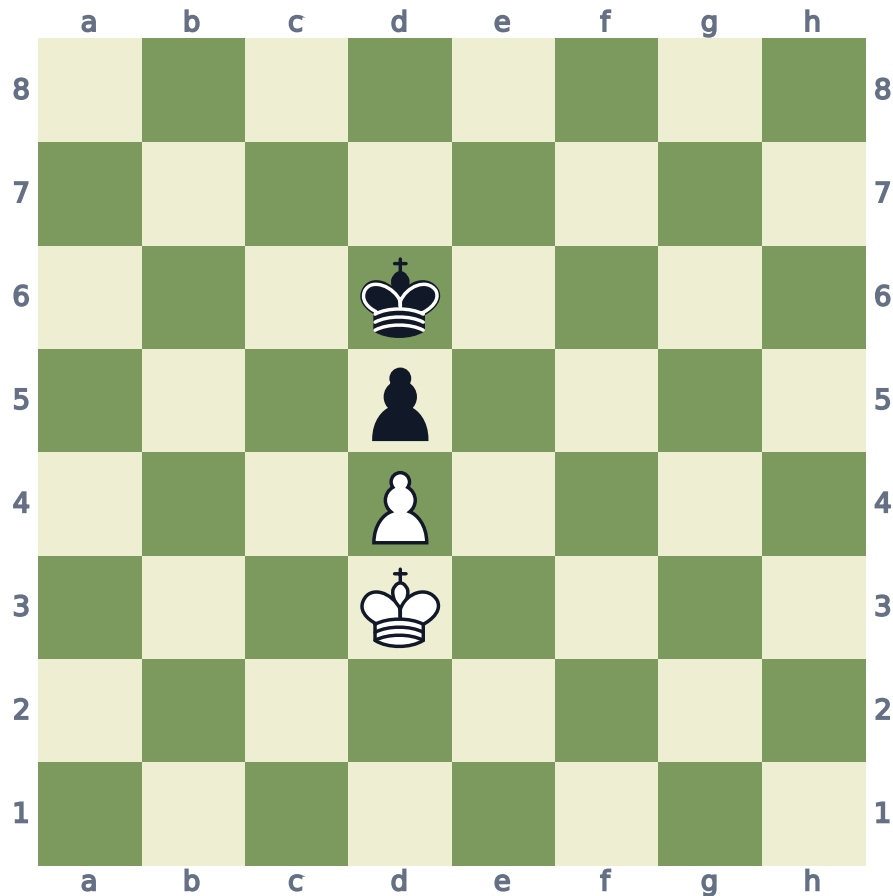
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 53

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Build the Fortress Around One Entry Point'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A fortress draws because the extra material cannot create a zugzwang, pawn break, or penetration. Identifying the single critical square allows a compact defense that never has to chase the attacker. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Build the Fortress Around One Entry Point'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 54

Coordinate Bishop and Knight for Mate

THE RULE

Drive the bare king toward the corner controlled by the bishop using a coordinated king, bishop, and knight net.

WHY IT WORKS

The bishop controls one color, the knight seals complementary escape squares, and the king compresses the box. The pieces cannot mate by isolated checks; they must reduce the king's area together.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

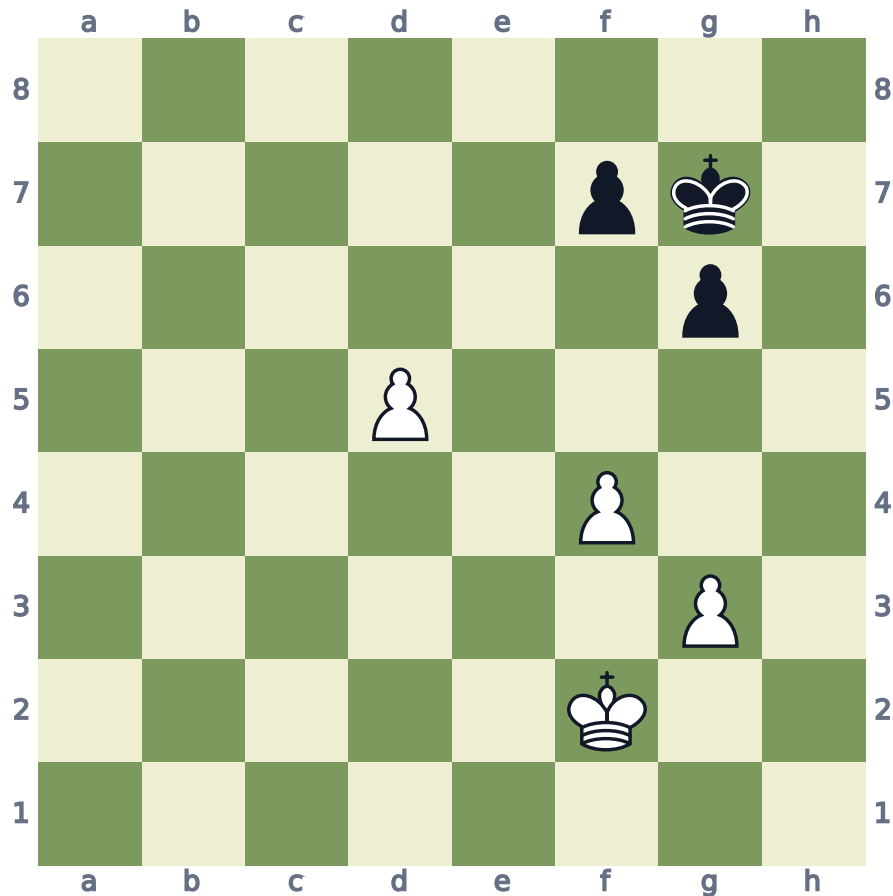
- Count material and passed pawns.
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- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 54

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Coordinate Bishop and Knight for Mate'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The bishop controls one color, the knight seals complementary escape squares, and the king compresses the box. The pieces cannot mate by isolated checks; they must reduce the king's area together. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Coordinate Bishop and Knight for Mate'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 55

Extinguish Counterplay Before Converting

THE RULE

In a winning ending, neutralize checks, passed pawns, and active pieces before collecting material.

WHY IT WORKS

The defender's chances come from activity, not static evaluation. Each source of counterplay removed makes the remaining calculation shorter and lowers the number of only moves required to win.

RECOGNITION CUES

a rook or queen must give checks; a king can be cut off; an exchange could enter a pawn ending; the defender still has active counterplay

APPLY IT OVER THE BOARD

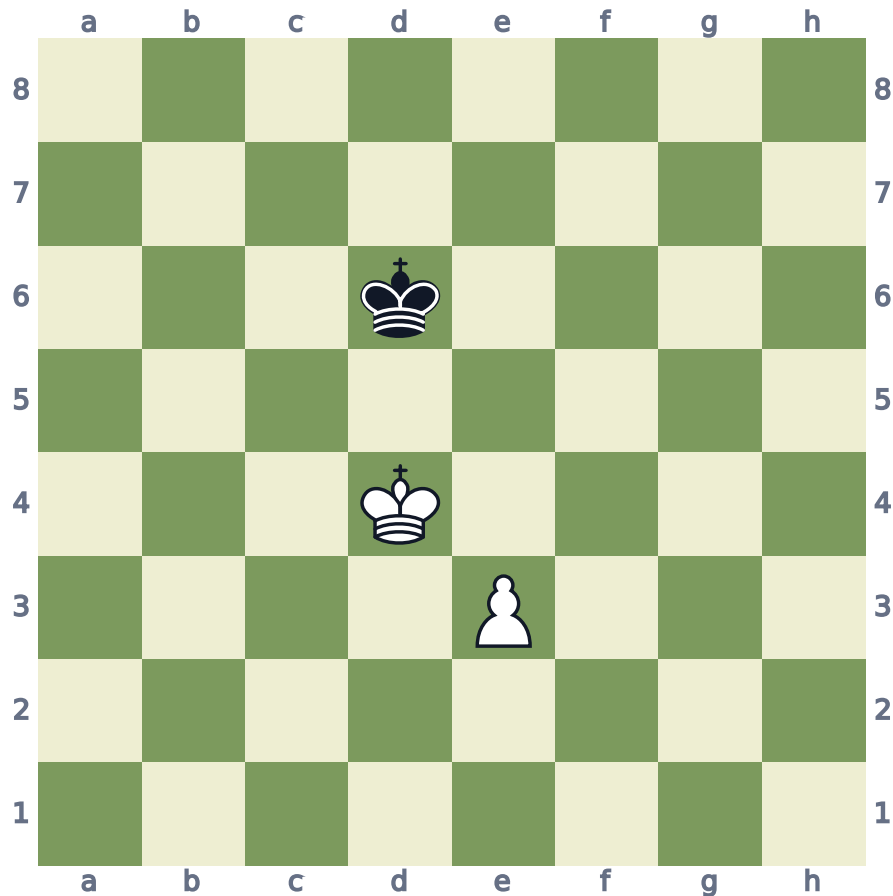
- Count material and passed pawns.
- Find the most active king and piece.
- Identify the exact target position.
- Calculate exchanges and pawn races completely.
- Remove counterplay before making routine progress.
-

BOUNDARY CONDITION

Do not simplify merely because you are ahead. Calculate the resulting pawn ending or fortress and preserve enough activity to answer checks and passed pawns.

BOARD LABORATORY 55

Use the idea, then try to disprove it



Exact endgame laboratory

YOUR TASK

Without moving the pieces, apply 'Extinguish Counterplay Before Converting'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The defender's chances come from activity, not static evaluation. Each source of counterplay removed makes the remaining calculation shorter and lowers the number of only moves required to win. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Extinguish Counterplay Before Converting'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

RETRIEVAL TEST

Can you explain the course without looking?

1

State the five-step decision protocol from memory.

2

Name three recognition cues that should make you slow down.

3

Choose five learning units and explain the causal mechanism behind each rule.

4

For each chosen rule, give one position where it should not be applied mechanically.

5

Describe the opponent's strongest practical resource in your most recent serious game.

6

Name one recurring error and the exact training drill that addresses its cause.

SCORING

2 points: accurate rule plus mechanism. 1 point: useful but incomplete. 0 points: slogan only. Restudy any item below 2 and retrieve it again tomorrow.

TRAINING PLAN

Four weeks to make the ideas usable

WEEK 1 - UNDERSTAND

Study the first quarter of the course. After every lesson, explain why the rule works and when it fails.

WEEK 2 - RECOGNIZE

Study the second quarter. Mix old and new diagrams so the name of the motif is not given in advance.

WEEK 3 - CALCULATE

Finish the course. Use a clock and write candidates, best defense, and endpoint evaluation.

WEEK 4 - TRANSFER

Play two slow games. Mark every moment when the course's protocol changed your decision, then analyze unaided.

LONG-TERM REVIEW

Return after 7, 14, and 30 days. Do not reread first. Attempt the retrieval test, review only missed items, and find one example from your own games.

REFERENCE

How this course fits the library

COURSE ROLE

Master the positions that must be known, while learning how to calculate unfamiliar endings from king activity, pawn races, checking geometry, and counterplay.

LIBRARY PRINCIPLE

Strong chess comes from a loop: notice the right features, generate serious candidates, calculate the opponent's best resistance, evaluate the destination position, and learn from the decision process.

ABOUT THE DIAGRAMS

Boards use Cburnett-style vector chess pieces with solid dark pieces and high-contrast olive and cream squares. Piece artwork is credited to Colin M. L. Burnett and the Lichess Cburnett set under GPLv2+.

EDITION NOTES

First edition, September 2026. 55 learning units. Explanations and exercises are designed for personal study and should be tested against actual games and concrete calculation.

THE FINAL QUESTION

What will you do differently on your very next move? If the course has not changed a behavior, choose one rule, attach it to a visible board cue, and practice it in five slow games.